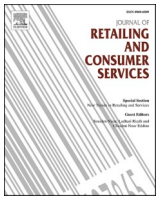
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Effect of infrastructure and technological factors on slum online commerce and product delivery: A structural functionalism perspective

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ABSTRACT

The global e-commerce market has surged, with Greater China and the U.S. generating over $3 trillion in combined revenue in 2023. In contrast, Sub-Saharan Africa (SSA) remains a marginal player, contributing less than 2 % to global e-commerce. This disparity is particularly evident in SSA's slum communities, which house over 60 % of the urban population. These slums face significant barriers—socioeconomic, infrastructural, and technological—that restrict their participation in online commerce and product delivery. While research has focused on health, sanitation, and poverty alleviation in SSA slums, the challenges of digital inclusion and integration into the global digital economy have been largely ignored. This study aims to address this gap by examining the infrastructural and technological factors that hinder online commerce and product delivery in SSA slums. Using a 5-point Likert scale and Partial Least Squares Structural Equation Modeling (PLS-SEM), the study reveals that while education and digital literacy and security concerns have no significant impact on slum online product delivery, factors like addressing systems, transportation infrastructure, telecommunications, and digital divide significantly influence online commerce and product delivery. Mediation analysis further indicated that enhancing product delivery systems can partially mitigate the adverse effects of poor infrastructure online commerce. The findings highlight the urgent need to improve physical infrastructure, including transportation and addressing systems, alongside technological solutions like telecommunications and digital divide. Without addressing these barriers, slum residents will remain disconnected from the digital economy, limiting their access to economic opportunities and essential goods and services.

1. Introduction

The digital economy has become a crucial driver of global economic growth, with online commerce playing a pivotal role in enhancing business expansion, financial inclusion, and economic participation (Hashemizadeh et al., 2023). In 2023, global e-commerce revenues surpassed $5 trillion, with China alone generating over $1.5 trillion due to its advanced digital infrastructure and logistics networks (Statista, 2023). Similarly, the United States and Europe continue to experience exponential growth in online commerce, benefiting from high internet penetration, efficient logistics, and strong regulatory frameworks (Statista, 2023). However, Sub-Saharan Africa (SSA) remains significantly behind in this digital transformation, contributing less than 2 % to global e-commerce revenues despite its growing urbanization and increasing mobile internet adoption (Statista, 2023). This digital divide is even more pronounced within SSA's informal settlements, where infrastructural deficiencies, technological limitations, and weak logistics networks severely hinder the expansion of online commerce and product delivery services (Takyi et al., 2021). Online commerce is increasingly recognized as a transformative tool for economic development in SSA, offering opportunities for businesses and individuals to access broader markets, reduce operational costs, and enhance financial inclusion (Yunus et al., 2022). Studies such as Abendin and Duan (2021) and Turkson et al. (2020) highlight how digital marketplaces are helping small businesses and entrepreneurs in SSA integrate into regional and global trade networks. Furthermore, effective product delivery systems are essential to sustaining this digital commerce ecosystem, as they ensure timely access to goods and enhance consumer trust (Morvey et al., 2024). However, the success of e-commerce in SSA largely depends on the availability of reliable infrastructure and technology, which remain major challenges, particularly in informal settlements. SSA slum settlements face severe infrastructural challenges that directly

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hinder online commerce and product delivery. Studies such as Philip and Williams (2019) and Frimpong et al. (2022) emphasize that poor road networks, unreliable electricity, and a lack of formal addressing systems make it difficult for e-commerce businesses to operate efficiently. The "last-mile delivery" problem which is a critical challenge in e-commerce logistics is significantly exacerbated by the absence of structured roads and clear delivery points (De Oliveira et al., 2019). This results in higher operational costs, delivery delays, and reduced consumer confidence, making e-commerce an unsustainable model for businesses in these environments. Furthermore, unreliable electricity affects both business operations and consumer engagement, as devices and internet access often depend on a consistent power supply (Ahmad and Zhang, 2021; Yan et al., 2013). On the technological front, SSA slums experience significant barriers that further restrict e-commerce growth. Limited access to affordable internet and digital devices creates a severe digital divide, excluding large portions of the population from online markets (Chen and Ruddy, 2023). Additionally, cybersecurity concerns pose major threats to both businesses and consumers in these areas. Radovanović et al. (2020) and Ferri et al. (2020) show that low levels of digital literacy prevent individuals from securely engaging in online transactions, making them vulnerable to cyber fraud. Moreover, e-commerce delivery personnel face heightened security risks, including theft and violence during transit, particularly in high-density informal settlements where weak law enforcement increases these dangers (Jamra et al., 2020).

Despite the rapid growth of online commerce and product delivery, and their potential to drive economic transformation, existing research on SSA slum communities has primarily focused on socio-economic deprivations such as sanitation, housing, health, and poverty alleviation, rather than on the infrastructural and technological barriers that hinder e-commerce growth. This leaves a critical gap in understanding how slum-specific infrastructural and digital deficiencies constrain online commerce and effective product delivery. Numerous studies have examined sanitation challenges in SSA slums (Haque et al., 2020; Katukiza et al., 2012), emphasizing the lack of clean water, inadequate waste disposal, and the spread of communicable diseases. While these issues are undeniably critical, they have been the primary research focus in slum environments, overshadowing the role of digital infrastructure in economic participation. Similarly, housing and settlement studies (Malik et al., 2020; Soyinka and Siu, 2018) have concentrated on precarious living conditions, land tenure issues, and eviction threats in SSA informal settlements. Although such research is valuable, it does not address how infrastructural and technological deficits impact the ability of slum residents to engage in online commerce and digital trade. Furthermore, the bulk of economic research on SSA slums has been framed around poverty reduction rather than digital economic participation (Razzaque and Zillur, 2018; Zulu et al., 2011). Studies in this domain have explored the effectiveness of microfinance (Hasan et al., 2022), social protection policies (de Haan, 2017), and informal labor dynamics (Amoah, 2024) in addressing urban poverty. However, they fail to analyze how e-commerce could serve as a catalyst for economic upliftment in slum economies or how infrastructural and technological bottlenecks limit this potential. The technological aspects of SSA's digital divide have also been extensively studied, but rarely in the context of slum-based e-commerce. For instance, research on SSA's digital economy (Kanyam et al., 2017; Myovella et al., 2020) has examined internet penetration rates, mobile money adoption, and e-government initiatives. However, these studies focus on national or urban-level digital transformations without addressing the unique constraints of informal settlements, where weak connectivity, digital illiteracy, and unreliable power supply exacerbate economic exclusion (Chen and Ruddy, 2023). Similarly, infrastructural research in SSA has focused on large-scale development projects such as highway expansion, urban planning, and electrification (Khanani et al., 2021; Saadaoui Mallek et al., 2024), but has neglected the specific infrastructural needs of slum residents for engaging in digital commerce. While transportation and logistics studies (Haji, 2021; Thiel, 2023) have analyzed the role of efficient road networks in fostering trade, they largely ignore the informal pathways, lack of structured addresses, and fragmented transport systems that characterize slum settlements. These are critical barriers that hinder last-mile delivery in slum-based e-commerce models (De Oliveira et al., 2019). Even within the limited body of research that addresses e-commerce and digital business in SSA, there is a tendency to focus on formal urban markets and high-growth sectors rather than slum-based commerce. For example, studies on fintech and digital trade in SSA (Osabutey and Jackson, 2024) highlight the success of mobile money services but do not explore how slum residents—many of whom lack stable internet access and digital skills—can effectively integrate into digital marketplaces. Moreover, while studies on gig economy platforms in SSA (Heeks et al., 2021) highlight the potential of digital labor, they fail to assess how logistical and technological limitations within slums impact participation in digital trade and e-commerce. This lack of attention to slum-specific digital barriers represents a major gap in the literature. While studies in developed economies have extensively linked e-commerce growth to advanced digital and logistical infrastructure (Haji, 2021; Thiel, 2023), there is a lack of research on how infrastructural and technological deficiencies uniquely impede e-commerce in SSA slums. Slum communities, which make up a large portion of SSA's urban population, remain largely excluded from e-commerce research despite their economic significance. Bridging these research gaps is critical for integrating SSA slum residents into the regional digital economy. Addressing infrastructural and technological challenges would not only enable more efficient online commerce and product delivery but also contribute to economic empowerment and financial inclusion for marginalized communities. Studies have shown that expanding digital trade and logistics in SSA could significantly boost annual e-commerce revenue and improve the socioeconomic standards of both consumers and businesses (Ahmedov, 2020; Kuteyi and Winkler, 2022). Moreover, enhancing access to digital markets can create new employment opportunities, increase household incomes, and reduce economic vulnerability in slum areas. This study, therefore, seeks to fill the existing gaps by addressing the following research questions.

- RQ1: How do infrastructure and technological factors influence online commerce and product delivery in SSA slum areas?
- RQ2: What roles do product delivery systems play in mitigating the effects of infrastructural deficits on slum online commerce?

By investigating these questions, the study aims to provide valuable insights into overcoming barriers to digital inclusion and unlocking the transformative potential of e-commerce in SSA slums. The findings will inform policymakers, technology providers, and e-commerce stakeholders on strategies for improving digital infrastructure and logistics in informal settlements, thereby fostering greater economic participation and sustainable development in SSA's digital economy.

2. Literature review

2.1. Barriers to online commerce and product delivery

The challenges facing online commerce (OCM) and product delivery (OPD) in Sub-Saharan African (SSA) slums are deeply rooted in infrastructural deficiencies, technological gaps, and socioeconomic inequalities, creating a stark contrast to more developed urban environments. These resource-deprived areas, as characterized by El-Haggar and Samaha (2019), lack the foundational elements necessary to support digital economic activities, hindering slum residents' participation in the rapidly expanding e-commerce ecosystem. Digital access which includes reliable internet connectivity, affordable devices, and digital literacy is essential for OCM and OPD, yet unequal digital penetration in SSA slums remains a significant barrier. Chen and Ruddy (2023) emphasize the transformative impact of digital connectivity,

citing Southeast and East Asia as prime examples where robust internet infrastructure has supported a dynamic e-commerce market valued at 72 billion in 2018, growing to over 100 billion by 2019. This underscores the critical role of digital infrastructure in driving economic performance. However, the persistent digital divide in SSA, as highlighted by Myovella et al. (2021), reveals that internet penetration remains disproportionately low, particularly in slum communities, limiting opportunities for e-commerce participation. Factors such as GDP per capita, electricity infrastructure, and political stability significantly influence internet access, with spillover effects between countries highlighting the need for regional collaboration to improve digital connectivity. The integration of infrastructure development with digital skill enhancement is critical for bridging the digital divide in slum environments. Chetty et al. (2018) argue that digital transformation necessitates a dual focus on infrastructure and skill development, as inadequate skills hinder residents' ability to access online platforms, comprehend product information, and complete transactions. Bejaković and Mrnjavac (2020) further emphasize that enhancing digital literacy not only influences individual economic participation but also drives broader socioeconomic development. Community-based initiatives, such as mobile digital literacy training units, are suggested as viable mechanisms to equip slum residents with the skills required to navigate the digital economy effectively. Reynolds et al. (2022) warn, however, that investments in digital technologies alone will not suffice if persistent digital divides in underdeveloped regions remain unaddressed, potentially negating the benefits of technological advancement. The digital divide in SSA slums is further compounded by socio-economic conditions, such as lower education levels, income disparities, and limited access to technology, as revealed by Öktem et al. (2021). Nevertheless, SSA's demographic advantage, with a youthful and digitally inclined population, presents an opportunity. According to the United Nations (2022), SSA has the youngest population globally, with over 60 % of urban youth residing in slums, as corroborated by Armah and Li (2023b). These digital natives have the potential to engage in e-commerce if barriers such as trust and security are addressed.

Trust and security are crucial for improving online engagement and participation in digital activities, yet slum residents often face heightened concerns about data privacy, cyber fraud, and the absence of robust legal protections. Noreen et al. (2021) argue that weak cybersecurity frameworks and high rates of cybercrime deter participation in e-commerce, as individuals perceive significant risks related to security, privacy, and financial transactions. However, the time and cost-saving benefits of mobile money underscore the importance of addressing these concerns. Chung et al. (2021) show the critical role of secure payment systems, strong encryption protocols, and community education in building trust in digital platforms. They argue that transparent platform policies and awareness campaigns can alleviate consumer concerns, enhancing confidence in e-commerce. In line with this, AI technologies have been leveraged to address privacy and security concerns, especially in social media platforms, by preventing data leakage. The study's findings show that AI-based privacy management, along with heightened user awareness of security measures, significantly reduces privacy concerns and enhances trust. When users feel confident about the protection of their data, they are more likely to engage with e-commerce platforms, thereby increasing adoption rates and fostering a secure online environment. Furthermore, platforms could enhance user trust by providing reliable dispute-resolution mechanisms, particularly in regions with minimal legal oversight. Payment systems which are one of the critical components of OCM and OPD, face significant challenges in slum contexts. According to Dadra et al. (2024), flexible and accessible payment options are vital for increasing consumer participation. However, cash remains the dominant payment method in slum areas due to limited access to digital payments and digital wallets. Although mobile payment systems like M-Pesa have gained traction in parts of SSA, their adoption in slum areas remains limited due to poor digital infrastructure and security concerns (Oliveira et al., 2016). Tounekti et al. (2020) highlight the potential benefits of promoting electronic payment adoption, including reduced dependency on cash, lower transaction costs, and enhanced convenience. Introducing hybrid payment models, such as cash-on-delivery integrated with digital platforms, and strengthening partnerships with local financial institutions could address these gaps and expand the reach of online commerce.

From a product delivery perspective, physical infrastructure such as inaccessible road networks and the absence of formal addressing systems remain significant challenges to efficient OPD in slums. Amaral and Cunha (2020) highlight the logistical challenges posed by the dense and informal nature of slums which increase last-mile delivery costs and reduce overall efficiency. The lack of proper infrastructure results in inefficient product distribution, with high transportation costs and delayed deliveries. These issues are exacerbated by the informal nature of settlements, which lack reliable address systems, making navigation and delivery processes challenging for e-commerce firms. Pongeluppe (2022) observes that e-commerce firms operating in informal settlements, such as Brazil's favelas, often face increased delivery costs and limited product availability due to a lack of investment in infrastructure. However, some firms have found success by investing in physical stores and engaging meaningfully with local communities, mitigating logistical challenges, and improving product accessibility. Similarly, Duarte et al. (2019) argue that traditional urban logistics models are often inadequate for slum environments, where logistical challenges such as poor infrastructure and informal settlements hinder efficient product delivery. Their study advocates for tailored solutions, including local warehouses and specialized delivery networks, to address these barriers and ensure effective product distribution. Mogire et al. (2022) further elaborate on the barriers faced by e-commerce in SSA slums, emphasizing the need for improved infrastructure and the integration of innovative technologies such as geo-location tools and mobile applications to enhance delivery efficiency. Reis et al. (2020) explore the role of technology in addressing last-mile delivery challenges, noting that innovative solutions such as drones, autonomous vehicles, and advanced mobile platforms hold great promise in reducing delivery times, increasing efficiency, and lowering costs, especially in areas with inadequate infrastructure.

2.2. *Emerging technologies in online commerce and product delivery*

The integration of emerging technologies has played a pivotal role in transforming the landscape of online commerce (OCM) and product delivery (OPD), significantly enhancing user experiences, operational efficiency, and logistics processes. However, the adoption and effectiveness of these technologies are often contingent upon the availability of robust physical and digital infrastructure and these are resources that are limited or absent in resource-constrained settings such as urban slums. This empirical review examines how emerging technologies are reshaping OCM and OPD, with particular focus on their potential benefits and limitations in under-resourced environments. Artificial Intelligence (AI) and Machine Learning (ML) have become central to the evolution of online commerce, enabling highly personalized consumer interactions and operational efficiencies. Rane et al. (2024) demonstrate how AI enhances customer experiences through tailored recommendations, while ML algorithms refine demand forecasting, inventory management, and pricing strategies, optimizing supply chain operations. In resource-rich contexts, these technologies have been credited with improving efficiency, reducing costs, and enhancing customer satisfaction (Zhang et al., 2023). However, in slum environments, these benefits are often inaccessible. The implementation of AI and ML in such settings is hindered by inadequate internet infrastructure, unreliable electricity, and low digital literacy rates among residents (Wolniak and Stecuła, 2024). As a result, the effectiveness of these technologies is constrained, and their adoption is slow, thus limiting the potential impact in these underserved areas. Similarly, Internet of Things (IoT) technology has revolutionized supply chain visibility by connecting devices to facilitate

real-time tracking and enhanced inventory control. Taj et al. (2023) highlight how IoT's capabilities in smart warehouses and logistics have transformed operational efficiency, making supply chains more agile and responsive. However, the reliance on stable internet connectivity and advanced hardware makes the implementation of IoT difficult in resource-constrained environments (Nilima et al., 2024). As IoT devices require constant connectivity and often rely on high-tech infrastructure, their adoption in slums is significantly limited. Inadequate network coverage and limited access to affordable smart devices present substantial barriers for residents, limiting their ability to participate in or benefit from these technological advancements. Blockchain technology, with its promise of decentralization, transparency, and enhanced security, is another emerging tool reshaping online commerce (Tricase et al., 2024). According to Dahal (2023), blockchain offers a secure environment for e-commerce transactions by reducing fraud, improving trust between consumers and merchants, and ensuring the authenticity of goods. In the delivery sector, blockchain enables secure payments and transparent transactions (Javaid et al., 2022). Despite its potential, the widespread implementation of blockchain faces challenges in low-resource settings. Dahal (2023) argues that the adoption of blockchain is hindered by limited access to digital payment systems, low levels of trust in new technologies, and the high costs associated with implementing blockchain infrastructure. In slums, where access to digital payment systems such as credit cards or mobile wallets is limited, the benefits of blockchain remain largely untapped.

In a last-mile delivery ecosystem, technologies like drones and autonomous vehicles are changing how goods are delivered to consumers. Zhang et al. (2023) and Garg et al. (2023) argue that drones can improve delivery speed, especially in congested urban areas, and reduce operational costs. Autonomous vehicles, as Çimen et al. (2024) discuss, are expected to enhance delivery speed and accuracy. However, these technologies face substantial challenges in slum settings. In areas with poorly maintained infrastructure, informal housing settlements, and regulatory hurdles, the viability of drone and autonomous vehicle deliveries is limited. The absence of reliable roads and transportation networks in slums presents significant barriers to the deployment of these advanced technologies (Sovacool et al., 2022). Furthermore, the high costs associated with developing and maintaining drone fleets or autonomous vehicles are often prohibitive in low-income areas. Augmented Reality (AR) and Virtual Reality (VR) are increasingly integrated into the online shopping experience, providing consumers with the ability to visualize products before making a purchase (Billewar et al., 2022). Zare Ebrahimabad et al. (2024) highlight that AR and VR enable consumers to try on products virtually or see them in their own living spaces, reducing uncertainty and enhancing satisfaction. These technologies have proven valuable in improving customer engagement and reducing return rates, particularly in well-resourced markets. However, in slum areas, the accessibility of AR and VR is severely limited. The high-speed internet required for seamless AR/VR experiences, combined with the need for advanced hardware such as VR headsets or AR-capable smartphones, makes these technologies largely inaccessible to residents in resource-constrained environments. Consequently, the widespread adoption of AR and VR for e-commerce in slums remains unlikely in the short term. 5G technology is set to transform online commerce by providing fast and low-delay communication as well as allowing consumers, merchants, and delivery services to interact in real-time (Kalkha et al., 2023). Mohdhar and Shaalan (2021) argue that 5G technology can streamline e-commerce experiences by allowing for faster data transfer and enhanced user interactions. However, the rollout of 5G networks is fraught with challenges in slum areas. The high cost of infrastructure development, coupled with the absence of existing telecommunications networks in many informal settlements, limits the ability of slum residents to access 5G services (Armah and Li, 2023a). Thus, while 5G has the potential to transform e-commerce in well-resourced areas, its benefits remain largely unrealized in underserved communities. Finally, robotics has revolutionized fulfillment centers and delivery systems. According to Ratchford et al. (2023), robots are increasingly used for sorting, packaging, and even delivering goods directly to customers, thereby enhancing efficiency and reducing labor costs. Robotics has proven successful in large-scale operations in wealthier regions, improving the speed and accuracy of logistics. However, in slums, the adoption of robotics is hindered by the high costs associated with automation, as well as the lack of skilled labor required to operate and maintain robotic systems. The absence of affordable technical training and the prohibitive costs of automation hardware means that the benefits of robotics are unlikely to reach low-income communities in the immediate future.

2.3. Global comparative studies in online commerce and product delivery

Comparative research on online commerce (OCM) and product delivery (OPD) across different regions provides critical insights into the diverse strategies, infrastructural challenges, and consumer behaviors shaping global e-commerce markets. Empirical studies underscore that while advanced economies rapidly integrate cutting-edge technologies into e-commerce operations, emerging markets contend with infrastructural deficits that slow down digital transformation. The global landscape of e-commerce is, therefore, highly uneven and influenced by varying levels of technological adoption, regulatory environments, and logistical challenges. A study by Hendricks and Mwapwele (2024) and Wang et al. (2022) confirms that developed regions such as North America and Europe have achieved high levels of e-commerce penetration due to robust digital infrastructure, widespread internet accessibility, and advanced payment systems. In these regions, artificial intelligence (AI), the Internet of Things (IoT), and blockchain have been seamlessly incorporated into e-commerce platforms to optimize inventory management, automate logistics, and enhance transaction security. The availability of high-speed internet infrastructure facilitates real-time data processing, ensuring seamless transactions and last-mile delivery efficiency. However, in emerging markets such as India, Brazil, and Sub-Saharan Africa (SSA), empirical evidence suggests that infrastructural limitations such as unreliable electricity, slow internet speeds, and underdeveloped logistical networks continue to hinder widespread e-commerce adoption (Hendricks and Mwapwele, 2024). This disparity underscores the need for region-specific technological solutions to bridge the digital divide in global e-commerce. Beyond infrastructure, payment systems also exhibit significant regional variations, influencing transaction efficiency and consumer trust. A study by Elverum Holm and Liu (2018) highlights stark contrasts in digital payment preferences across different economies. In the United States and Europe, traditional banking systems have cemented the dominance of credit cards and digital wallets like PayPal and Apple Pay. Conversely, China has seen an overwhelming shift toward mobile-first payment platforms such as Alipay and WeChat Pay, largely bypassing credit card reliance. This divergence is attributed to differences in financial system evolution, consumer preferences, and regulatory policies governing digital payments. The findings suggest that businesses seeking to expand into global markets must tailor their payment systems to align with regional financial behaviors, ensuring seamless integration with consumer preferences.

Last-mile delivery, a critical determinant of e-commerce success, also varies widely across geographical regions. Research by Mnyakin (2023) illustrates that in the United States, vast geographic distances pose logistical challenges for timely deliveries. To address these inefficiencies, companies have increasingly adopted localized distribution centers, autonomous delivery vehicles, and drone technology to optimize last-mile logistics. In contrast, Japan's densely populated urban centers and sophisticated logistics networks enable more efficient delivery models. Studies by Mohsen (2024) indicate that Japanese companies leverage smart routing algorithms and eco-friendly electric-powered delivery vehicles to minimize congestion and lower costs. These findings emphasize that the effectiveness of last-mile

delivery strategies is contingent on urban planning, population density, and infrastructural readiness. In emerging markets, where informal settlements and unstructured addresses present logistical hurdles, companies must explore alternative delivery solutions, such as mobile pickup stations and crowd-sourced courier services, to navigate these challenges. Regulatory frameworks further contribute to disparities in global e-commerce operations, particularly concerning data privacy and security policies. Hoofnagle et al. (2019) provide empirical evidence that in the European Union, stringent regulations such as the General Data Protection Regulation (GDPR) impose strict compliance requirements on businesses regarding data collection and customer privacy. While these policies enhance consumer trust and data security, they also increase operational costs for businesses that must invest in compliance mechanisms. In contrast, research by Wong (2020) reveals that countries in Southeast Asia operate under relatively lenient data protection laws, enabling businesses to engage in aggressive data-driven marketing and personalized advertising with fewer restrictions. While this regulatory flexibility allows for faster technological deployment, it raises ethical concerns regarding data privacy and consumer vulnerability to digital exploitation. Thus, businesses operating in multiple regions must navigate regulatory landscapes carefully, balancing compliance with competitive advantage. Consumer behavior remains another fundamental driver of regional differences in e-commerce adoption and delivery preferences. Nguyen et al. (2019) found that Western consumers prioritize speed, convenience, and cost-effectiveness, leading to widespread demand for rapid delivery services, price optimization algorithms, and automated customer service solutions. By contrast, empirical studies on the Middle Eastern market indicate a strong preference for luxury goods, high-touch customer service, and personalized shopping experiences. This cultural divergence necessitates region-specific marketing and service strategies to align with local consumer expectations. For instance, while U.S.-based retailers focus on same-day delivery and price competition, e-commerce platforms in the Middle East emphasize premium packaging, concierge delivery services, and exclusive membership benefits to attract high-net-worth customers.

2.4. Technology adoption theories

This section reviews various theoretical frameworks pivotal to understanding and modeling online commerce and product delivery. Key models include the Technology Acceptance Model (TAM), Unified Theory of Acceptance and Use of Technology (UTAUT), Theory of Planned Behavior (TPB), Diffusion of Innovation Theory (DOI), Last-Mile Theory, and the SERVQUAL model. These frameworks collectively address technology adoption, consumer behavior, and service quality, offering valuable insights into the dynamics of e-commerce and logistics systems. TAM, introduced by Davis (1989), emphasizes perceived usefulness (PU) and perceived ease of use (PEOU) as critical determinants of technology adoption. In e-commerce, these constructs explain consumer decisions to adopt systems like mobile apps, online payment gateways, and delivery tracking technologies. Studies by Yaqub et al. (2024) and Tseng et al. (2022) confirm that PU and PEOU significantly influence user engagement and satisfaction, highlighting the model's utility in assessing customer preferences in technology-driven retail. Over time, TAM has evolved through extensions like TAM2 (Venkatesh and Davis, 2000) and UTAUT, incorporating additional social and contextual factors to enhance predictive accuracy. The UTAUT framework, developed by Venkatesh et al. (2003), builds on TAM by integrating constructs like performance expectancy, effort expectancy, social influence, and facilitating conditions, moderated by demographic and experiential variables. Its application in online commerce reveals that consumer adoption of e-commerce platforms often hinges on perceived benefits, ease of use, and social pressures. For instance, studies by Escobar-Rodríguez and Carvajal-Trujillo (2014) and Belanche et al. (2020) demonstrate the model's relevance in analyzing customer intentions to adopt real-time tracking apps and mobile payment systems, which are integral to modern delivery services. Ajzen's Theory of Planned Behavior (TPB) (1991) further complements these insights by emphasizing attitudes, subjective norms, and perceived behavioral control as predictors of consumer behavior. TPB's application in e-commerce explains why factors like user-friendly interfaces, peer influence, and control over delivery options enhance adoption rates. Empirical research by Tang et al. (2021) and Leon et al. (2023) highlights the importance of these dimensions, with perceived control over delivery features significantly influencing customer satisfaction and repeat usage in e-commerce settings. DOI, developed by Rogers et al. (2014), provides a lens to understand how innovations diffuse within a population. Its constructs—relative advantage, compatibility, complexity, trialability, and observability—offer a framework to analyze the adoption of emerging e-commerce platforms and logistics technologies. Studies by Alam et al. (2008) reveal that perceived benefits and infrastructure compatibility are vital to encouraging consumer adoption of e-commerce services. DOI's focus on adopter categories, from innovators to laggards, also aids in crafting targeted strategies to accelerate adoption cycles. Last-Mile Theory addresses the logistical challenges inherent in the final stage of product delivery, emphasizing efficiency and cost-effectiveness. In the context of e-commerce, this theory helps optimize delivery networks, particularly in urban and underserved areas. Belanche et al. (2020) highlight the role of last-mile innovations like real-time tracking in improving consumer satisfaction, underscoring the theory's relevance to e-commerce logistics. Finally, the SERVQUAL model measures service quality through dimensions like reliability, responsiveness, and empathy, offering a framework to assess and enhance consumer experiences. Studies by Zeithaml et al. (1988) and Bogatzki et al. (2020) demonstrate how SERVQUAL applies to online commerce, with reliable delivery and responsive customer support emerging as key factors in customer satisfaction. Together, these theories provide a multidimensional understanding of online commerce and product delivery, addressing both technological and behavioral aspects. While TAM and UTAUT focus on adoption drivers, TPB and DOI offer insights into behavioral intentions and innovation diffusion. Last-mile theory and SERVQUAL complement these by addressing logistical efficiency and service quality, ensuring a holistic approach to e-commerce research. These frameworks continue to guide scholarly and practical efforts to enhance digital commerce systems and optimize logistics in evolving markets.

2.5. Structural functionalism as a framework for technology

Structural functionalism, a foundational sociological perspective rooted in the works of Émile Durkheim and later advanced by Talcott Parsons and Robert Merton, conceptualizes society as a system of interconnected structures that function collectively to maintain stability (Merton, 1957; Parsons and Mayhew, 1982). Each societal component contributes to equilibrium, and any dysfunction can disrupt social order (Ormerod, 2020). While traditionally applied to institutions such as family, education, and governance, scholars have increasingly explored its relevance in the realm of technology. In particular, the structural functionalist lens provides insights into how technological advancements integrate with and shape societal frameworks, reinforcing stability while also introducing potential dysfunctions. Technology, within this framework, functions as an evolving subsystem that enhances productivity, communication, and societal coordination. Luhmann (1995) conceptualizes technology as an adaptive structure that facilitates social cohesion and economic efficiency. One of the clearest examples of this integration is in communication technology, where digital platforms such as social media and instant messaging serve the critical function of sustaining information flow and reinforcing societal interconnectedness (Can, 2021). The widespread adoption of these technologies illustrates how digital networks have become integral to maintaining social stability by enabling continuous interaction and engagement among

individuals and institutions. Beyond communication, structural functionalism also offers a compelling explanation of technology's role in industrial and economic systems. Ismagilova et al. (2019) highlight how technological advancements optimize production processes and labor distribution, directly contributing to economic stability. Automation in manufacturing, for instance, reduces human error and increases efficiency, ensuring that economic structures remain robust and sustainable (ElMaraghy et al., 2021). Similarly, digital financial systems, such as online banking and mobile payment platforms, play a crucial role in maintaining economic equilibrium by facilitating seamless transactions and fostering economic inclusion (Ferrari, 2022). These technological systems function as stabilizing agents that sustain financial infrastructure and ensure the continuous flow of economic activity.

However, structural functionalism does not merely emphasize the positive contributions of technology; it also sheds light on the dysfunctions that arise from technological evolution. Merton (1957) concept of latent and manifest functions is particularly relevant in this context, as it helps explain the unintended consequences of technological advancements. While automation enhances industrial efficiency (a manifest function), it simultaneously leads to job displacement and economic disparities (a latent dysfunction). Similarly, digital communication technologies, while promoting connectivity, have also facilitated issues such as cyberbullying and misinformation, which disrupt social cohesion (Midigo, 2024). These examples illustrate how technological progress can simultaneously reinforce and undermine societal structures, necessitating careful regulation and adaptation to mitigate unintended consequences. In governance, structural functionalism underscores the significance of technology in maintaining social order and administrative efficiency. E-governance initiatives, including digital identity systems and automated public services, have emerged as critical tools for streamlining bureaucratic functions and enhancing citizen engagement (Addo and Senyo, 2021). By digitizing governmental processes, these technologies contribute to institutional stability and efficiency. However, as Heeks (2022) notes, digital governance systems can also exacerbate systemic inequalities if access is unevenly distributed, reinforcing a digital divide that marginalizes certain populations. This paradox highlights the dual role of technology in governance—while it fosters efficiency, it also necessitates proactive measures to ensure equitable accessibility. Structural functionalism provides a valuable theoretical framework for understanding the integration of technology within societal structures. It highlights both the functional contributions and potential dysfunctions of technological systems, offering insights into how societies can harness technological advancements to maintain stability while mitigating their disruptive effects. As technology continues to evolve, applying structural functionalism remains crucial in assessing its societal impact and ensuring that technological developments serve the collective good. Extending this perspective, structural functionalism offers a novel lens for examining online commerce (OCM) and product delivery (OPD) in slum environments. While existing research predominantly focuses on family, education, governance economic, infrastructural, and behavioral challenges, few studies integrate a systemic sociological approach. This study pioneers the application of structural functionalism in this domain, presenting an innovative framework that aligns with the interconnected nature of OCM and OPD. Given that slum economies operate through adaptive informal networks, the functionalist approach provides an ideal theoretical lens for understanding how infrastructure and technology interact to sustain OCM and OPD. By framing these elements as interdependent structures within an evolving system, this research extends structural functionalism into an underexplored yet highly relevant technological context, offering new insights into the role of technology in facilitating economic and social resilience.

3. Theoretical framework and modeling

3.1. Adopted theoretical framework

Structural functionalism as highlighted in section 2.5, views society as a complex system in which various components interact to maintain stability and equilibrium (Izadi et al., 2020). It emphasizes the interdependence of societal elements such as institutions, norms, and values (Turner, 2001). Widely applied in fields such as urban planning and economic development (Kingsbury and Scanzoni, 2009; Neef, 2018), structural functionalism offers a robust framework for understanding the challenges of OCM and OPD in SSA slums. Unlike traditional technology adoption theories, which focus on individual adoption behaviors, structural functionalism examines systemic issues, making it more suitable for analyzing e-commerce in environments with infrastructural deficiencies such as unreliable telecommunications (UTC), poor transportation (TPI), and inadequate addressing systems (ADS). Additionally, technological barriers including the digital divide (DTD), low education and digital literacy (EDL), and security concerns (STC) further hinder e-commerce participation in slums, reinforcing the need for a systemic approach. This study incorporates key derivatives of structural functionalism such as Parsons' AGIL framework, Merton's manifest and latent functions, and Luhmann's systems theory, to strengthen its applicability. Parsons' AGIL framework (Adaptation, Goal Attainment, Integration, and Latency) provides a structured approach to analyzing how e-commerce ecosystems function despite infrastructural constraints (Parsons and Mayhew, 1982). Adaptation (A) explains how slum economies compensate for structural limitations by using informal addressing systems and mobile payment solutions. Goal Attainment (G) highlights entrepreneurial strategies tailored to slum environments. Integration (I) focuses on resolving technological and social barriers, such as closing the digital divide and improving cybersecurity. Latency (L) accounts for deep-rooted informal economic structures and cultural norms, demonstrating that e-commerce models must align with socio-economic realities (Irfan, 2023). Merton (1957) concept of manifest and latent functions further supports the relevance of structural functionalism in OCM and OPD. For instance, the manifest function of OPD is to facilitate product delivery, while its latent function is to mitigate infrastructural failures through adaptive logistics such as motorcycle couriers, decentralized distribution hubs, and AI-powered tracking systems. Additionally, Luhmann (1995) systems theory emphasizes that trust is a macro-structural issue rather than an individual psychological decision. In SSA slums, trust deficits arising from cybersecurity concerns, unreliable telecommunications, and financial fraud risks impede digital payment adoption and online transactions. Structural functionalism, through these extensions, highlights systemic adaptation, stability, and interdependence, reinforcing its suitability for modeling e-commerce in slum settings.

In contrast, traditional technology adoption theories, including TAM, UTAUT, TPB, DOI, Last-Mile Theory, and SERVQUAL focus on individual adoption behaviors, consumer attitudes, and service quality (as outlined in Section 2.4). TAM and UTAUT emphasize perceived usefulness, ease of use, and social influence as key drivers of e-commerce adoption (Malatji et al., 2020; Rejali et al., 2023). TPB explores attitudes, subjective norms, and perceived control, shedding light on how peer influence and interface design shape consumer decisions (Ajzen, 1991). DOI examines the diffusion of innovations, highlighting compatibility and trialability as critical factors in e-commerce adoption (Duarte Alonso et al., 2024). The last-mile theory addresses logistical inefficiencies, optimizing delivery services, while SERVQUAL assesses service quality based on dimensions such as reliability and responsiveness (Jiang et al., 2020; Zygiaris et al., 2022). While these theories provide valuable insights for improving digital commerce platforms and logistics, they assume stable infrastructure, making them inadequate for slum environments. For example, TAM and UTAUT's (Yaqub et al., 2024) emphasis on perceived usefulness and ease of use becomes

irrelevant in slums, where inadequate addressing systems and unreliable transport infrastructure make last-mile delivery nearly impossible. Structural Functionalism's Adaptation function provides a more contextually relevant perspective by explaining how slum economies compensate for addressing system failures through GPS-based informal addressing, digital mapping services, and community reference points which are strategies that traditional adoption models overlook. Similarly, DOI's assumption of predictable innovation diffusion is disrupted in slum environments (Das, 2022), where transportation constraints hinder product movement. Structural Functionalism's Integration function offers a more realistic framework by illustrating how decentralized logistics systems including motorcycle-based deliveries, informal package relay networks, and community-driven distribution hubs compensate for transport deficiencies.

Beyond infrastructure, technological and economic barriers further limit the applicability of traditional adoption models. TPB and DOI assume equal access to digital payment systems (de Luna et al., 2019), failing to account for high transaction costs, outdated devices, and digital illiteracy which are common barriers in SSA slums. TPB emphasizes perceived control, while DOI focuses on innovation diffusion, but both fail to address how socio-economic constraints influence digital payment adoption. Structural Functionalism's Adaptation function provides a more practical lens by explaining how alternative payment models, such as cash-on-delivery and mobile money intermediaries, enable e-commerce participation despite technological limitations. Additionally, Luhmann's systems theory reframes trust as a macro-structural issue, emphasizing how weak cybersecurity frameworks and inadequate consumer protection laws hinder digital transactions—factors overlooked by TAM and UTAUT, which assume that ease of use and perceived usefulness drive adoption. Furthermore, Last-Mile Theory and SERVQUAL assume formalized logistics systems, failing to account for the role of informal networks in slum economies. Structural Functionalism's Latency function recognizes these deep-rooted informal structures, emphasizing that e-commerce models must align with existing socioeconomic dynamics rather than impose external frameworks (Irfan, 2023). While traditional technology adoption theories provide valuable micro-level insights into consumer behavior, they lack the macro-level explanatory power needed to address the infrastructural and technological constraints of slum e-commerce. Structural functionalism offers a comprehensive systems-based approach that considers the interdependencies between infrastructure, technology, and social structures. By accounting for the gaps beyond traditional technology adoption theories, structural functionalism provides a more realistic and effective framework for analyzing and improving OCM and OPD in SSA slums.

3.2. Conceptual framework

Fig. 1 illustrates the conceptual framework of the study, emphasizing the impact of infrastructural and technological barriers on Online Commerce (OCM) and Online Product Delivery (OPD) within Sub-Saharan Africa (SSA) slum communities. Grounded in Structural Functionalism, the framework posits that the growth and success of OCM and OPD depend on the effective functioning of infrastructure—Addressing Systems (ADS), Transportation Infrastructure (TPI), and Telecommunication (UTC)—and a robust technological ecosystem that bridges the Digital Divide (DTD), promotes Education and Digital Literacy (EDL), and mitigates Security Concerns (STC) among slum residents. The framework highlights the synergistic interactions between these factors. ADS and TPI improve logistical efficiency, enabling accurate and timely product deliveries, while TC and the absence of DTD facilitate seamless communication and trustworthy online transactions. EDL empowers slum residents to navigate and utilize e-commerce platforms effectively, and STC addresses security concerns, fostering confidence in online transactions. Together, these elements ensure the reliability, accessibility, and scalability of OCM and OPD, promoting digital inclusion and economic participation in SSA slums. Conversely, dysfunction in any of these subsystems destabilizes the broader ecosystem. For instance, inadequate ADS and TPI can lead to delivery delays, unreliable TC and DTD can hinder communication and accessibility, and limited EDL and weak STC can reduce user engagement and trust. Research confirms that slums face significant infrastructural and technological barriers (El-Haggar and Samaha, 2019; Takyi et al., 2021), which exacerbate challenges for OCM and OPD. This study's conceptual framework examines these impacts directly and explores whether OPD can mitigate the negative effects of infrastructural challenges on OCM. The subsequent sections define each construct in detail and provide their operational context, setting the foundation for hypothesis development and statistical analysis to validate the framework's propositions.

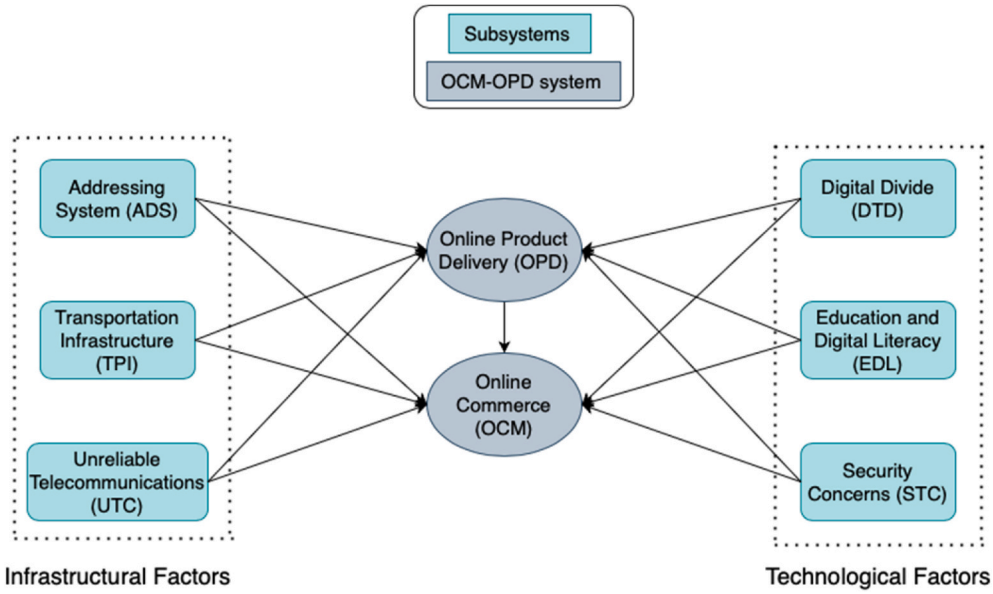


Fig. 1. Conceptual framework of infrastructure and technology on OCM and OPD.

3.3. Explanation of key constructs

Online commerce (OCM) refers to the buying and selling of goods and services through digital platforms, which can significantly enhance economic participation in slum communities. The construct is measured through various dimensions such as accessibility, convenience, product range, and user-friendliness. For instance, Excellent Mofokeng and K Tan (2021) found that platforms being accessible and offering a wide range of suitable products are key factors in enabling economic participation. Further, user-friendliness ensures engagement by minimizing digital barriers. A practical example could be slum residents using mobile apps for daily essentials, which demonstrates the role of OCM in creating opportunities for economic diversification and resilience, thus enhancing the community's overall stability (Cordes and Marinova, 2023).

Online product delivery (OPD): OPD refers to the logistics and distribution processes involved in delivering purchased items to customers. Slum communities face challenges related to reliability, customer satisfaction, and delivery infrastructure. OPD in the study's context is measured by factors like service reliability, customer satisfaction, and the effectiveness of tracking systems. Research by De Oliveira et al. (2019) and Praet and Martens (2020) highlights that reliable delivery services, particularly those incorporating innovative solutions such as collection points and real-time tracking, significantly improve the delivery experience in slums. These factors enhance customer confidence, reinforcing the interdependence between online commerce and product delivery in fostering economic development and social cohesion.

The addressing system (ADS) construct defines the infrastructure that supports the identification and delivery of goods to specific locations, which is crucial for efficient online product delivery (OPD). This construct highlights challenges such as unreliable addressing systems, difficulty in providing precise address information, and the struggle of delivery agents to locate residences in slum areas. According to Abebrese (2019), slums often lack formal or consistent addressing systems, leading to delays and inefficiencies in product delivery. For example, a delivery agent may be unable to find a slum-based customer due to poorly marked or non-existent addresses, delaying the transaction and reducing customer satisfaction. Thiel (2023) emphasizes that the absence of reliable addressing systems hampers timely deliveries, further impeding the effectiveness of online commerce in such regions. Addressing these challenges could significantly improve OPD performance and e-commerce outcomes.

Transportation infrastructure (TPI) defines the state of transport systems essential for online product delivery (OPD). This construct addresses challenges like poor road conditions, unreliable transport services, and delays in delivery caused by inadequate infrastructure. According to Banerjee et al. (2020) and Meersman and Nazemzadeh (2017), efficient transportation systems are vital for timely deliveries and e-commerce growth. However, slum communities often face disrepair and congestion, as highlighted by Mohmand et al. (2017), limiting the primary function of transportation, which is to move goods efficiently, and thus disrupting its secondary function of facilitating commerce. For example, a slum-based vendor might experience frequent delivery delays due to impassable roads, leading to customer dissatisfaction and reduced trust in online platforms. By improving transportation infrastructure, as emphasized by Mesjasz-Lech and Włodarczyk (2022), delivery efficiency and OCM participation can significantly increase.

Unreliable telecommunication (UTC) refers to the inefficiencies in communication infrastructure that disrupt the seamless operation of online commerce and product delivery (OCM and OPD). This construct encompasses challenges such as unstable internet connectivity, slow internet speeds, and unreliable telecommunication services, which hinder essential activities like order placement, route optimization, and vendor communication. For instance, Donou-Adonsou et al. (2016) highlight how poor connectivity in underdeveloped regions negatively affects e-commerce engagement, while Okyere et al. (2018) emphasize the difficulty of using navigation and tracking apps due to unreliable networks. For instance, a slum-based business might struggle to coordinate deliveries because of dropped calls or delayed online updates, causing inefficiencies and customer dissatisfaction.

The digital divide (DTD) construct captures the disparities in internet access, availability of digital technologies, and digital literacy that hinder participation in online commerce and product delivery (OCM) within slum communities. This study highlights key dimensions of the digital divide, including limited internet access, inadequate digital infrastructure, difficulties in navigating online platforms, and unequal access to the benefits of e-commerce. As noted by Wamuyu (2017) and Valarezo et al. (2020), these disparities significantly restrict individuals' ability to engage in online transactions or compete with businesses in more connected regions, thereby exacerbating economic inequality. The DTD construct offers a comprehensive framework to analyze how technological inequities create barriers to inclusive digital participation, highlighting the importance of bridging these gaps and fostering equitable access to e-commerce opportunities.

The education and digital literacy (EDL) construct captures the critical role of education and digital skills in enabling participation in online commerce and product delivery (OCM and OPD). Drawing on Structural Functionalism, education serves as a key subsystem, equipping individuals with the necessary skills to effectively engage in economic activities. In slum communities, however, low levels of digital literacy and inadequate educational initiatives create significant barriers to participation. For instance, Mahmood et al. (2022) and Anita et al. (2021) highlight that residents in such areas often lack awareness of internet platforms, struggle with basic digital skills, and face limited access to educational programs targeting computer literacy. The EDL construct evaluates these challenges through items like the scarcity of digital skills and inadequate initiatives.

Security concerns (STC) capture the fears and risks associated with online transactions and product delivery in slum areas. This construct includes concerns over personal information security, risks of online fraud, and physical safety threats to delivery personnel in densely populated environments. Ahmed et al. (2024) discuss how inadequate cybersecurity infrastructure in underprivileged areas increases distrust, preventing residents from fully engaging in online commerce. For instance, slum residents may hesitate to shop online due to fear of their financial data being compromised or fraudulent activities by unverified sellers. Similarly, delivery personnel may face theft risks while navigating high-crime areas.

3.4. Structural equation modeling (SEM) of the framework

3.4.1. Definition of constructs and variables

Dependent Variables (System)—OCM: Online Commerce and OPD: Online Product Delivery (Mediator).

Independent Variables (Subsystems): *Infrastructural Factors (INF)*—Addressing System (ADS), Transportation Infrastructure (TPI), Unreliable Telecommunication (UTC). *Technological Factors (TECH)*—Digital Divide (DTD), Education and Digital Literacy (EDL), Security Concerns (STC).

Latent Constructs: ***Infrastructure (INF)***—ADS, TPI, UTC. ***Technology (TECH)***—DTD, EDL, STC.

The hypothesized Pathways of the model are formulated as follow:

Direct effects of Infrastructure (INF) and Technology (TECH) on OCM (Online Commerce):

$$OCM = f(INF, TECH) \tag{3.1}$$

Direct effects of Infrastructure (INF) and Technology (TECH) on OPD (Online Product Delivery):

$$OPD = f(INF, TECH) \tag{3.2}$$

Mediating effect of OPD on the relationship between INF, TECH and OCM:

$$OCM = f(OPD, INF, TECH) \tag{3.3}$$

3.4.2. Structural equations

The measurement model (latent constructs) for the SEM are express as follow:

For INF (Infrastructure):

$$INF = \lambda_1 ADS + \lambda_2 TPI + \lambda_3 UTC + \varepsilon_{INF} \tag{3.4}$$

Where λ_1, λ_2, λ_3 is the factor loadings that represent the contribution of each indicator (ADS, TPI, UTC) to the latent construct INF and ε_{INF} is the measurement error or unexplained variance in the latent variable INF.

For TECH (Technology):

$$TECH = \lambda_4 DTD + \lambda_5 EDL + \lambda_6 STC + \varepsilon_{TECH} \tag{3.5}$$

Where: λ_4, λ_5, λ_6 is the factor loadings that represent the contribution of each indicator (DTD, EDL, STC) to the latent construct TECH and ε_{TECH} is the measurement error or unexplained variance in the latent variable TECH.

For the structural model, the effect of INF and TECH on OPD can be express as:

$$OPD = \alpha_0 + \alpha_1\, INF + \alpha_2 TECH + \varepsilon_{OPD} \tag{3.6}$$

Where: α_0 is the intercept term, representing the baseline level of OPD when INF and TECH are zero. α_1 and α_2 are path coefficients, representing the direct effects of INF and TECH on OPD, respectively. ε_{OPD} is residual variance, accounting for the portion of OPD unexplained by INF and TECH.

Similarly, the effect of INF, TECH, OPD on OCM can be presented as:

$$OCM = \beta_0 + \beta_1\, INF + \beta_2 TECH + \beta_3 OPD + \varepsilon_{OCM} \tag{3.7}$$

Where: β_0 **is intercept term**, representing the baseline level of OCM when INF, TECH and OPD are zero. $\beta_1, \beta_2, \beta_3$ are p**ath coefficients**. β_1 is direct effect of INF on OCM, β_2 is direct effect of TECH on OCM. β_3 is direct effect of OPD on OCM. ε_{OCM} is **residual variance**, accounting for the portion of OCM unexplained by INF, TECH, and OPD.

To examine the total, direct, and indirect effects, the relationships below can be assessed.

$$\text{Direct Effects} \begin{cases} \text{INF}\rightarrow\text{OCM}\ (\beta_1) \\ \text{TECH}\rightarrow\text{OCM}\ (\beta_2) \\ \text{INF}\rightarrow\text{OPD}\ (\alpha_1), \\ \text{TECH}\rightarrow\text{OPD}\ (\alpha_2) \\ \text{OPD}\rightarrow\text{OCM}\ (\beta_3) \end{cases} \tag{3.8}$$

$$Indirect\ Effect = (\alpha_1 \times \beta_3) + (\alpha_2 \times \beta_3) \tag{3.9}$$

$$Total\ Effect = \beta_1 + \beta_2 + Indirect\ Effect \tag{3.10}$$

3.5. Hypotheses development

Slums, characterized by inadequate infrastructure, limited access to technology, and high population density (Takyi et al., 2021), present unique challenges to online commerce and product delivery. As digital platforms increasingly shape commerce, these areas are often marginalized in terms of logistics and connectivity. From a **Structural Functionalist perspective** (Izadi et al., 2020), which views society as a system of interconnected parts working together to maintain stability and functionality, both infrastructure and technology play critical roles in facilitating or hindering the development of slum economies. This framework helps explain how improvements in infrastructure and technological factors can enhance efficiency in slum commerce and delivery systems, ensuring that these communities are integrated into the broader economy.

3.5.1. The effect of online product delivery (OPD) on online commerce (OCM)

In slum environments, OPD demonstrates the potential to overcome infrastructure and technological barriers, enabling e-commerce growth. Despite challenges such as unreliable addressing systems, poor transportation infrastructure, and limited digital access, OPD leverages localized innovations and adaptive delivery methods, such as motorcycle couriers, drone package delivery, and hybrid tracking solutions, to ensure the timely delivery of goods (Benarbia and Kyamakya, 2021; Excellent Mofokeng and K Tan, 2021). This resilience fosters trust in e-commerce platforms and promotes customer loyalty, even in resource-constrained settings. The success of OPD in navigating these challenges directly enhances OCM by addressing the critical "last-mile" gap that often deters slum customer participation. Studies like Cabeza-Ramírez et al. (2022) and Vakulenko et al. (2019) have shown that reliable delivery improves customer satisfaction, stimulates repeat purchases, and encourages informal economies to integrate into formal digital ecosystems. Furthermore, innovative OPD strategies in slum areas, such as real-time delivery updates and flexible drop-off points, build confidence in digital platforms, contributing to OCM's accessibility and inclusivity. Thus, OPD's adaptability and operational effectiveness catalyze OCM growth, transforming logistical challenges into opportunities for expanding market reach and fostering digital inclusion in underserved communities. In light of this, the study posits that.

H1. OPD has a positive impact on OCM.

3.5.2. The effect of addressing system (ADS) on OCM and OPD

From the lens of structural functionalism, the addressing system (ADS) plays a foundational role in supporting the structural integrity of OCM and OPD. However, in slum environments, where inadequate urban planning results in informal and inconsistent ADS, this subsystem fails to function effectively, disrupting the stability of the broader digital commerce ecosystem (Oloo, 2018; Solomon and van Klyton, 2020). The absence of formal ADS creates significant barriers to OCM by undermining logistical efficiency and customer trust. In slum communities, residents often rely on landmarks or descriptive directions, leading to delivery delays, misrouted parcels, and increased operational costs for logistics companies (Dumedah, 2021; Owusu-Adjei, 2019). This logistical inefficiency discourages e-commerce platforms from targeting these areas, further marginalizing slum populations from participating in the digital economy. The lack of reliable addressing also increases the digital divide, reducing the accessibility and inclusivity of online commerce for underserved communities (Frimpong et al., 2022). Similarly, OPD relies on efficient last-mile delivery mechanisms, which are severely constrained by the absence of formal ADS. Delivery personnel must navigate irregular layouts and unmarked streets, often resorting to local knowledge and trial-and-error methods, increasing delivery times and costs (Dumedah and Eshun, 2020; Frimpong et al., 2022). These inefficiencies erode customer satisfaction and deter future transactions, undermining the role of OPD as a critical component for OCM growth. Structural functionalism emphasizes that when key subsystems like ADS fail, the entire system—comprising OCM and OPD—suffers disequilibrium. Therefore, addressing this critical infrastructure gap is essential for promoting economic inclusion in slum communities. Therefore, the study proposes the following hypotheses.

H1a. ADS has a significant adverse impact on OCM.

H2a. ADS has a significant adverse impact on OPD.

3.5.3. The effect of transportation infrastructure (TPI) on OCM and OPD

Transportation infrastructure (TPI) serves as a vital enabler of OCM and OPD. However, considering the nature of SSA slums, inadequate TPI

presents significant barriers that hinder the growth and functionality of the OCM-OPD system. According to Amaral and Cunha (2020), poor road conditions, unpaved and narrow pathways, and impassable road networks characterize slum areas, making smooth goods mobility exceptionally challenging. These infrastructural deficiencies lead to delays, increased delivery costs, and unreliable service, ultimately diminishing the attractiveness of e-commerce for slum residents and businesses (Katsela et al., 2022). For OPD, the adverse effects of poor TPI are most pronounced. Delivery vehicles frequently struggle to navigate congested and poorly maintained roads, causing inefficiencies and higher fuel consumption. In many cases, delivery personnel must resort to non-motorized transport or on-foot deliveries, which are slower and less reliable (Comi and Savchenko, 2021). Such logistical challenges reduce the reliability of OPD services and strain the operational budgets of logistics providers, limiting their willingness to serve slum communities. These transportation challenges also ripple into OCM, where poor TPI discourages e-commerce vendors from expanding into slum regions due to high delivery costs and frequent delays. Consumers in these areas, in turn, face limited access to e-commerce services, reinforcing economic exclusion and widening the digital divide (Liu et al., 2024). Structural functionalism highlights these disruptions as critical to understanding the systemic imbalance in slum e-commerce ecosystems. Based on these effects, the study posits that.

H1b. TPI has a significant adverse impact on OCM.

H2b. TPI has a significant adverse impact on OPD.

3.5.4. The effect of unreliable telecommunication (UTC) on OCM and OPD

In slum communities, telecommunication infrastructure is not merely a channel for digital transactions but a critical enabler of systemic integration between consumers, suppliers, and logistics networks. However, unreliable telecommunication (UTC) disrupts these interactions by creating gaps in essential digital functions. Challenges such as unstable internet connectivity, slow speeds, and intermittent mobile signals make it difficult for residents in SSA slums to engage with online commerce platforms or participate in seamless product delivery systems (Mishra et al., 2022; Tounekti et al., 2020). These limitations compromise the structural functionality of OCM and OPD, leading to inefficiencies and diminished trust in digital solutions. For OCM, poor telecommunication reliability hinders users from accessing e-commerce services consistently, resulting in incomplete transactions, canceled orders, and frustration among potential customers. This affects platform usability and erodes confidence in the digital economy (Chen, 2020). In OPD, UTC aggravates logistical inefficiencies by causing communication delays, misaligned deliveries, and real-time tracking failures. Such disruptions raise operational costs and undermine the reliability of delivery services, further discouraging vendor participation in slum markets (Onunga and Mbugua, 2021). The structural functionalism perspective highlights how these telecommunication deficiencies destabilize the interdependencies within slum-based digital ecosystems, perpetuating exclusion and hindering economic progress (Nguimkeu and Okou, 2021).

H1c. UTC has a significant adverse impact on OCM.

H2c. UTC has a significant adverse impact on OPD.

3.5.5. The effect of digital divide (DTD) on OCM and OPD

The digital divide (DTD) represents a critical barrier to the integration of slum communities into the digital economy, particularly in SSA. This divide stems from unequal access to internet connectivity, digital devices, and digital literacy, limiting the ability of slum residents to participate in online commerce and product delivery systems. In the study's context, the digital divide disrupts the equilibrium of interdependent societal subsystems, including economic, technological, and logistical networks. In the context of OCM, limited internet access and inadequate digital infrastructure restrict residents' ability to browse online platforms, engage in e-commerce transactions, and benefit from digital financial services (Philip et al., 2017; Wamuyu, 2017). Challenges such as navigating poorly optimized platforms and lacking familiarity with e-commerce tools further widen the gap, excluding slum populations from opportunities for economic growth and perpetuating inequality (Nguyen and Oliver, 2023). For OPD, DTD increases logistical inefficiencies. Inconsistent internet access and communication delays prevent real-time order tracking, transaction confirmations, and timely delivery updates. These issues lead to missed or delayed deliveries, undermining trust in digital solutions and reducing the effectiveness of product delivery systems in slum environments (Armah and Li, 2023a). Structural functionalism emphasizes the adverse ripple effects of the digital divide across these interconnected systems, highlighting its role in reinforcing socioeconomic disparities.

H1d. DTD has a significant adverse impact on OCM.

H2d. DTD has a significant adverse impact on OPD.

3.5.6. The effect of education and digital literacy (EDL) on OCM and OPD

In the digital age, education and digital literacy (EDL) serve as the foundation for engaging with modern commerce systems. However, slum communities in SSA experience a severe deficit in these critical skills, which undermines their ability to participate in OCM and benefit from OPD. This inadequacy stems from challenges, including the scarcity of formal educational institutions, insufficient digital training initiatives, and limited exposure to e-commerce platforms. From a structural-functionalism standpoint, the societal subsystem responsible for education fails to perform its function, creating an effect that affects the broader economic and technological systems. For OCM, the lack of EDL translates into limited knowledge of online marketplaces and hesitation to engage with digital systems. Many residents are unable to navigate e-commerce platforms, leading to a diminished user base and restricted market growth (James and Jeffrey James, 2021). This not only excludes individuals from accessing goods and services but also discourages businesses from investing in digital infrastructure targeting these areas. In terms of OPD, the absence of digital competence affects communication between customers and delivery providers. Misunderstandings during order placement or difficulties in using digital tracking tools result in delayed deliveries and logistical inefficiencies (Jagtap et al., 2020). These challenges increase operational costs for service providers and affect customer trust which further excludes slum residents from the digital economy. Bridging the EDL gap is essential to fostering equitable participation in OCM and OPD, enhancing economic inclusion, and enabling smoother integration of slum communities into digital commerce ecosystems. From this context, the study hypothesized that.

H1e. EDL has a significant adverse impact on OCM.

H2e. EDL has a significant adverse impact on OPD.

3.5.7. The effect of security concerns (STC) on OCM and OPD

In SSA slum communities, security concerns (STC) are a critical hindrance to the successful integration of OCM and OPD systems. From the perspective of structural functionalism, the malfunctioning of societal structures, such as digital infrastructure and physical safety measures, influences the balance necessary for efficient economic transactions. The security challenges in these areas manifest in several forms: the vulnerability of personal data in online transactions, the potential for online fraud, and the physical risks faced by delivery personnel navigating overcrowded and poorly constructed environments. Slum areas often lack adequate infrastructure for securing digital interactions, hence, fostering a climate of fear around online purchases. Consumers may hesitate to participate in e-commerce due to concerns about privacy violations, fraud, and unreliable payment systems, which ultimately limits the expansion of OCM (Irshad and Arnisha, 2020). In addition, the high incidence of criminal activity in these communities

increases apprehension about the safety of delivery operations. Narrow streets, informal addressing systems, and lack of security measures create further challenges for delivery personnel, who are frequently targets of theft or harm (Armah and Li, 2023b). This insecurity not only undermines customer confidence but also impedes the efficient movement of goods, damaging the viability of OPD systems. Given these pressing concerns, the study proposes the following hypotheses.

H1f. STC has a significant adverse impact on OCM.

H2f. STC has a significant adverse impact on OPD.

3.5.8. The mediating effect of OPD as a mitigator

From a structural-functionalism perspective, online product delivery (OPD) plays a critical role in addressing systemic deficiencies—such as inadequate addressing systems (ADS), poor transportation infrastructure (TPI), unreliable telecommunications (UTC), and security concerns (STC)—that hinder the adoption and success of online commerce (OCM) in Sub-Saharan African (SSA) slum environments. These interconnected challenges, if unaddressed, disrupt the equilibrium of the socioeconomic structures necessary for effective digital transactions. OPD, however, serves as a functional solution, bridging gaps and enabling slum communities to participate in the digital economy. First, the absence of formal addressing systems in SSA slums poses significant obstacles to online commerce by creating inefficiencies in product delivery. Without standardized address formats, delivery agents often face delays and increased operational costs. OPD systems equipped with digital tools, such as GPS-enabled localization and mobile mapping technologies, offer innovative solutions by enabling precise customer identification and route optimization (Habjan et al., 2014; Harris et al., 2015). These technologies mitigate ADS-related challenges by creating virtual addressing systems that bypass the reliance on formal infrastructure, ensuring efficient and reliable deliveries. Second, poor transportation infrastructure in slums characterized by narrow pathways, unpaved roads, and high congestion complicates logistics. OPD systems adapt to these conditions by employing alternative delivery methods, such as motorcycles, bicycles (Armah and Li, 2023a), or drone couriers (Liu, 2019), which are more agile and capable of navigating such environments. These last-mile delivery strategies not only reduce delays but also enhance the overall reliability of the delivery process, ensuring customer satisfaction even in the face of infrastructural inadequacies. Third, unreliable telecommunications infrastructure in slums limits communication between customers and businesses. This disrupts the coordination required for successful e-commerce transactions. OPD systems address these challenges through real-time delivery tracking, which reduces the dependency on direct communication while increasing transparency and trust. Customers can independently monitor their orders, predict arrival times, and receive notifications, thus overcoming UTC limitations. Lastly, security concerns such as theft, fraud, and physical risks to delivery personnel represent critical barriers to OCM in slums. OPD systems mitigate these risks by implementing secure delivery practices, such as tamper-proof packaging, designated safe delivery zones, and digital proof-of-delivery protocols. These measures protect both customers and delivery agents, fostering a safer environment for e-commerce activities (Marta et al., 2023; Neama et al., 2016). By effectively addressing these structural deficits, OPD systems not only enhance the efficiency of OCM but also build consumer trust and operational resilience. The adaptive capacity of OPD to mitigate challenges related to ADS, TPI, UTC, and STC makes it a critical enabler of digital inclusion and economic growth in SSA slums. Consequently, the study posits the following hypotheses.

H3a. OPD mediates the relationship between ADS and OCM.

H3b. OPD mediates the relationship between TPI and OCM.

H3c. OPD mediates the relationship between UTC and OCM.

H3d. OPD mediates the relationship between STC and OCM.

4. Methodology

4.1. Population and sample size

The study focuses on slum communities in Sub-Saharan Africa (SSA), which are home to a significant portion of the region's population. Given the large and diverse nature of these communities, obtaining accurate population statistics is essential for determining the minimum sample size required to draw reliable inferences and for guiding the questionnaire distribution and data collection processes. However, conducting research in slum areas is challenging due to the lack of comprehensive and consistent population data. Estimates of slum populations in SSA vary widely across countries and regions, making it difficult to establish precise figures. For instance, Li et al. (2023) reported that approximately 218 million Africans lived in SSA slums in 2020, while the Daniel (2015) estimated that 238 million people in SSA resided in slums in 2018. Using these two data points, the study calculated a population growth rate of −8.4 %, indicating a declining trend. Assuming this rate of decline continues, the projected slum population in SSA for 2024 was estimated to be approximately 182 million. To determine the appropriate sample size for the study, the following statistical parameters were applied: n (sample size), N (population size), e (margin of error, set at 0.05), P (population proportion, set at 0.5), and Z (Z-score, set at 1.96 for a 95 % confidence level). Based on these parameters, the minimum required sample size was calculated to be 384 respondents. While a sample size larger than 384 was deemed acceptable, the study adopted this figure as the minimum target for each selected slum community. A total of 14 slum communities were included in the study, with data correspondents stationed in each community to facilitate questionnaire distribution. In total, 6,300 questionnaires were distributed, averaging approximately 450 questionnaires per community.

The study employed an online-based method to administer the questionnaires, which were designed with a "required option" feature to prevent missing data. This feature ensured that respondents could not submit incomplete questionnaires, thereby addressing the issue of missing data and improving data quality. Of the 6,300 questionnaires distributed, 5,830 were fully completed and returned, representing a response rate of 92.5 %. This high response rate is particularly noteworthy in the context of contemporary research trends. For example, Holtom et al. (2022) observed an upward trend in response rates across various fields, including e-commerce and supply chain management. Their analysis of 1,014 surveys from 703 articles published between 2010 and 2020 found that average response rates increased from 48 % in 2005 to 68 % in 2020. Against this backdrop, the 92.5 % response rate achieved in this study is considered exceptional. To further ensure the accuracy, reliability, and validity of the data, the study implemented rigorous quality control measures at both the pre-and post-data collection stages. Pre-quality control measures were applied during the data collection phase to minimize errors and inconsistencies, while post-quality control measures were used to verify the credibility, objectivity, and generalizability of the findings to slum communities across Sub-Saharan Africa. These processes are discussed in greater detail in subsequent sections of the study.

4.2. Sampling techniques and slum community selection

The study adopted a mixed-method sampling approach, combining purposive and simple random sampling techniques to ensure a balanced representation of both specific and context-rich insights as well as broader and generalizable findings. This approach was designed to address the unique challenges of conducting research in slum communities while maintaining methodological rigor and ensuring the reliability of results. Purposive sampling was employed as a pre-quality control measure to select slum communities that were particularly relevant to the study's objectives. Selection criteria focused on identifying slum areas in SSA that faced significant infrastructural and

technological challenges, particularly those impeding online commerce and product delivery. The rationale for selecting these specific slum communities was grounded in their significant economic activities, which are often informal, population density, and the extent to which they experienced challenges such as unreliable addressing systems, inadequate transportation infrastructure, poor telecommunication networks, and limited digital connectivity. For example, slums like Nima (Ghana), Ajegunle (Nigeria), and Mukuru Kwa Njenga (Kenya) were selected due to their high population density, informal economic activities, and limited addressing systems. Similarly, Kibera (Kenya), Old Fadama (Ghana), and Kimicanga (Rwanda) were included as they are among the largest slums in Africa, characterized by inadequate addressing systems, unreliable telecommunications, and poor transport infrastructure. Other slums, such as Dutse Alhaji (Nigeria), Vingunguti (Tanzania), and Kroo Bay (Sierra Leone), were chosen for their poor digital connectivity, high unemployment rates, and limited access to financial services. Additionally, slums like Manzese (Tanzania), Bé Lagoon (Togo), and Merkato Slum (Ethiopia) were included due to their weak logistics, supply chain inefficiencies, and limited e-commerce adoption. Finally, Mathare (Kenya) and Briqueterie (Cameroon) were selected for their high crime rates, digital illiteracy, and language barriers in digital commerce. These factors were critical for understanding how infrastructural and technological challenges impact online commerce and product delivery within these communities. By including a diverse range of slums with varying barriers, the study ensured that the data and findings were representative of the broader challenges faced by slum residents and businesses in SSA.

Following the purposive selection of communities, simple random sampling was another pre-quality control measure rigorously implemented to select individual respondents within each identified slum. This method ensured that every individual had an equal probability of being chosen, thereby minimizing selection bias, ensuring sample representation of the broader slum population, and enhancing the reliability of the results. The randomization process was executed by trained data collection assistants, who were residents of the respective regions and had an in-depth understanding of local dynamics. These assistants facilitated the distribution of online questionnaires and ensured adherence to fair and transparent randomization procedures. To precisely implement the randomization process, the study followed a structured three-step procedure. First, data collection assistants used a systematic approach to distribute the survey links randomly within community-specific social media groups, messaging apps (e.g., WhatsApp, Telegram), marketplaces, and digital platforms frequented by slum residents. Second, respondents were encouraged to participate voluntarily, and responses were monitored to prevent multiple submissions from the same individuals, thereby ensuring randomization integrity. Third, time-based randomization was implemented, where questionnaires were sent out at different times of the day to capture a diverse range of respondents, including those engaged in informal employment or with limited online access at specific hours. To address digital exclusion, data assistants offered their smartphones to randomly selected offline volunteers, enabling participation from those without personal devices. This method ensured a fair distribution of participants while maintaining the randomness of selection. To uphold the integrity of the data collection process, several pre-quality control measures were instituted. Data assistants received training on ethical considerations, including obtaining informed consent and maintaining respondent confidentiality. Additionally, the online questionnaires were designed with a "required option" feature to prevent incomplete responses, addressing the common challenge of missing data in survey-based research. These methodological strategies from the purposive selection of communities to the rigorous randomization of respondents and the implementation of quality control measures significantly strengthened the study's ability to draw meaningful and generalizable conclusions. By integrating these approaches, the study generated accurate, reliable, and actionable insights into the barriers to online commerce and product delivery in SSA slums, ensuring the credibility and applicability of the findings across similar contexts in the region.

4.3. Variables operationalization and instrumentation

The study employs a 5-point Likert scale, ranging from 1 (strongly disagree) to 5 (strongly agree), to measure respondents' levels of agreement with various statements. The Likert scale is a well-established method for capturing respondents' attitudes, perceptions, and behaviors across diverse domains, including infrastructure, technology adoption, and online commerce (Joshi et al., 2015). Its application in this context is supported by its ability to measure latent constructs—such as perceptions of infrastructural deficiencies and technological barriers—with precision and comparability. The study identifies key constraints hindering the growth of online commerce (OCM) and online product delivery (OPD) as being rooted in a lack of infrastructure and technological barriers in slum environments. From these broad challenges, the study focuses on specific constructs, including poor addressing systems (ADS), inadequate transportation infrastructure (TPI), unreliable telecommunication services (UTC), the digital divide (DTD), limited education and digital literacy (EDL), and security concerns (STC).

The items associated with these constructs were pretested through an iterative process of item development, which included a pilot test conducted in three waves. In each wave, items with lower factor loadings were replaced. Additionally, expert reviews were conducted to ensure construct validity. Experts' assessments and feedback were used to refine the appropriateness and clarity of the questions for the data collection exercise, as outlined by Armah and Li (2023a). These processes are pre-quality control measures to ensure the construct validity of the primary questionnaire. Each construct was operationalized using structural functionalism frameworks (Ormerod, 2020; Parsons and Mayhew, 1982), enabling the study to conceptualize and define the constructs in alignment with its theoretical foundation. For example: ADS items evaluate the reliability and effectiveness of addressing systems in slum areas, focusing on their role in locating residences and ensuring timely online product deliveries. TPI items capture the impact of poor transportation conditions and infrastructure on the efficiency and timeliness of product deliveries. UTC items reflect challenges arising from unstable and inefficient telecommunication services, including internet connectivity and communication issues, in supporting online commerce and delivery. DTD items measure the extent to which slum residents face difficulties accessing and utilizing digital technologies and online platforms, which hinders equal participation in online commerce. EDL items assess the influence of limited education and digital literacy on residents' ability to engage effectively with online commerce platforms and enhance their technological skills. STC items measure perceived risks of fraud, data theft, and physical insecurity in slum environments, which deter participation in online shopping and delivery activities. OCM items evaluate the accessibility, convenience, product variety, and user-friendliness of e-commerce platforms for slum residents. OPD items assess the reliability, efficiency, and customer satisfaction with delivery services, including tracking systems and collection points in slum communities.

The reliability of the Likert scale-based instrument was assessed using factor loadings (FL), Cronbach's Alpha (α), composite reliability (CR), average variance extracted (AVE), discriminant validity (DV), and the standardized root mean square residual (SRMR). The results indicated that FL values exceeded 0.708, while Cronbach's Alpha (α) and CR were both above 0.700, confirming internal consistency. The SRMR value (0.078) was below the 0.08 threshold, indicating a well-fitting model. Additionally, AVE explained more than 50 % of the variance, confirming strong convergent validity. A detailed breakdown of these assessments is provided in Tables 4–6. Overall, the findings demonstrate the robustness and reliability of the measurement model.

4.4. Demographic distributions

The demographic data of the study reveals a sample primarily composed of young, self-employed males with varying levels of education and employment experience. The gender distribution shows a significant male majority (68.24 %) compared to females (31.76 %), which aligns with studies highlighting gender disparities in digital access due to socioeconomic and cultural barriers (Shiferaw, 2024). The age distribution indicates that 47.27 % of respondents are between 18 and 28 years old, while 33.49 % fall within the 29–39 age bracket, reflecting Africa's predominantly youthful population, as noted by the United Nations (2022) This age demographic is particularly relevant to digital commerce, as younger individuals are generally more tech-savvy and open to online business models. Educational attainment among respondents shows promise for digital literacy, with 30.89 % having completed secondary school and 29.04 % holding diplomas, suggesting a foundation for integrating slum residents into the digital economy. However, nearly half (49.33 %) have only a basic or secondary education, which may limit their capacity to fully engage in e-commerce. Employment data further highlights the entrepreneurial nature of the sample, with 44.10 % engaged in self-employment, a trend commonly observed in informal economies (Bird et al., 2022). However, the 22.13 % unemployment rate and the 15.02 % of respondents with no work experience indicate significant barriers to economic inclusion. Internet awareness is universal (100 %), and usage patterns show high engagement, with 48.89 % accessing the Internet more than seven times daily. Despite this, 65.49 % spend less than 30 min per session online, hinting at infrastructural limitations such as poor connectivity or device constraints. Commerce behavior indicates that 46.49 % engage in both online and offline shopping, while 39.65 % prefer offline shopping, likely due to accessibility challenges. Social networks play a pivotal role in ICT adoption, with 59.50 % of respondents relying on peer influence, supporting previous research that highlights the role of community networks in digital technology dissemination (Gwaka et al., 2022). The representativeness of the sample is justified by its alignment with broader demographic and socioeconomic trends in Sub-Saharan African slums, particularly in terms of youth dominance, self-employment reliance, and infrastructural constraints, which are well-documented factors influencing digital inclusion and e-commerce engagement in developing economies. Consequently, while the sample effectively captures the realities of digital commerce participation in SSA slums, interventions must address education, employment, and infrastructure challenges to ensure equitable digital economic integration. These insights formed the basis for understanding the broader impact of infrastructural and technological barriers on online commerce and product delivery (OCM and OPD) in SSA slums. The demographic characteristics of the respondents are detailed in Table 1, and the survey items, along with their operationalization for each construct, are provided in Table 9 in the Appendix.

Table 1
Respondents' demographic distribution.

Demographic features	Frequency	Percentage (%)
Gender		
Male	3977	68.24
Female	1853	31.76
Age		
18–28	2757	47.27
29–39	1952	33.49
40–50	867	14.88
51–61	108	1.85
62 and above	146	2.50
Level of education		
Basic	1074	18.44
Secondary	1801	30.89
Diploma	1694	29.04
Bachelor's	839	14.42
Master's	422	7.26
Ph.D.	–	–
Employment status		
Public sector	512	8.79
Private sector	1461	25.06
Self-employment	2569	44.10
Currently unemployed	1288	22.13
Employment duration		
No experience	875	15.02
1–5	2633	45.22
6–10	1158	19.86
11–15	957	16.43
16+	207	3.55
Internet awareness		
Yes	5830	100.00
No	–	–
Internet usage		
1-2 times per day	424	7.28
3-4 times per day	1189	20.41
5-6 times per day	1366	23.42
7+ times per day	2851	48.89
Surfing reason		
Lifestyle	2944	50.54
Information	509	8.74
Business	1622	27.82
Purchasing	755	12.96
Commerce		
Online shopping	807	13.86
Offline shopping	2312	39.65
Both activities	2711	46.49
Surfing duration		
< 30 mins	3811	65.49
60 min	1132	19.42
90 min	614	10.54
120+ mins	273	4.69
ICT recommendations		
Family	385	6.61
Friends	3468	59.50
Colleagues	1029	17.66
Traditional media	948	16.29

4.5. Data analysis technique

The impact of infrastructure and technological factors on online commerce (OCM) and product delivery (OPD) in Sub-Saharan African (SSA) slums remains largely unexplored, with significant gaps in the literature. This study aims to bridge this gap by assessing: (1) the direct impact of OPD on OCM, (2) the effects of ADS, TPI, UTC, DTD, EDL, and STC on OCM, (3) their effects on OPD, and (4) the mediating role of OPD in these relationships. Given the lack of empirical validation in slum-based e-commerce and logistics management, a robust analytical framework is essential. To evaluate these relationships, the study employs Partial Least Squares Structural Equation Modeling (PLS-SEM) rather than Covariance-Based Structural Equation Modeling (CB-SEM) due to its suitability for complex, exploratory research with predictive objectives (Hair et al., 2019; Sarstedt et al., 2021). PLS-SEM efficiently handles models with multiple constructs and intricate interactions, making it ideal for analyzing the systemic barriers to OCM and OPD in SSA slum communities. Unlike CB-SEM, PLS-SEM is also robust to non-normal data distributions which is a crucial advantage given the heterogeneity of slum populations. To further validate the suitability of this method, normality tests using the Kolmogorov–Smirnov and Shapiro–Wilk tests were conducted, yielding p-values below 0.05, confirming the non-normality of the data and reinforcing the appropriateness of PLS-SEM for hypothesis testing. According to Hair et al. (2014), the required sample size in PLS-SEM should be ten times the number of arrows pointing to any given variable. With 13 arrows, a minimum of 130 valid responses is required. The study's sample size of 5,830 respondents far exceeds this threshold, ensuring statistical power, significance, and generalizability. Additionally, while the study benefits from a large dataset, PLS-SEM remains advantageous for future research with smaller sample sizes. To ensure methodological rigor, diagnostic tests confirmed no multicollinearity issues (VIF <5). The R^2 values

Table 2
Results of the measurement model and descriptive statistics of the items.

Construct items	λ	∝	ρA	CR	AVE	VIF	Mean	Std
ADS1	0.830	0.836	0.841	0.891	0.672	2.239	3.858	0.965
ADS2	0.836					2.225	3.895	0.961
ADS3	0.867					2.302	3.789	0.915
ADS4	0.742					1.709	3.703	1.126
DTD1	0.757	0.815	0.864	0.873	0.633	2.724	3.649	1.105
DTD2	0.716					2.429	3.563	1.148
DTD3	0.847					2.282	3.494	1.153
DTD4	0.853					2.043	3.515	1.137
EDL1	0.620	0.816	0.865	0.878	0.647	1.398	3.592	1.022
EDL2	0.800					1.739	4.144	0.924
EDL3	0.914					3.669	3.895	0.751
EDL4	0.853					3.112	3.897	0.748
OCM1	0.921	0.881	0.887	0.919	0.739	3.479	4.042	0.909
OCM2	0.878					2.729	4.087	1.046
OCM3	0.856					2.317	4.073	1.030
OCM4	0.776					1.725	4.054	0.949
OPD1	0.905	0.827	0.852	0.886	0.661	2.586	3.992	0.795
OPD2	0.820					1.970	3.961	0.881
OPD3	0.776					1.660	3.805	0.932
OPD4	0.742					1.522	3.838	0.921
STC1	0.822	0.803	0.804	0.872	0.629	2.175	3.619	1.014
STC2	0.798					1.684	4.124	0.913
STC3	0.764					1.579	3.882	0.775
STC4	0.788					2.013	3.502	0.859
TPI1	0.919	0.878	0.885	0.917	0.735	3.446	4.027	0.890
TPI2	0.878					2.644	4.069	0.976
TPI3	0.854					2.242	4.003	1.032
TPI4	0.773					1.715	4.011	0.945
UTC1	0.853	0.768	0.777	0.852	0.592	1.905	3.984	0.788
UTC2	0.781					1.605	3.963	0.868
UTC3	0.721					1.377	3.782	0.930
UTC4	0.714					1.372	3.794	0.953

Table 3
Discriminant validity using the Fornell-Larcker criterion.

	ADS	DTD	EDL	OCM	OPD	STC	TPI	UTC
ADS	0.820							
DTD	0.509	0.795						
EDL	0.195	0.201	0.804					
OCM	0.322	0.245	0.362	0.860				
OPD	0.263	0.124	0.411	0.353	0.813			
STC	0.335	0.358	0.691	0.422	0.304	0.793		
TPI	0.290	0.210	0.293	0.707	0.297	0.397	0.857	
UTC	0.259	0.155	0.494	0.388	0.718	0.359	0.322	0.769

Table 4
Discriminant validity using Heterotrait–Monotrait (HTMT).

	ADS	DTD	EDL	OCM	OPD	STC	TPI	UTC
ADS								
DTD	0.631							
EDL	0.232	0.230						
OCM	0.375	0.276	0.415					
OPD	0.316	0.128	0.476	0.416				
STC	0.409	0.428	0.876	0.495	0.369			
TPI	0.337	0.241	0.334	0.802	0.350	0.466		
UTC	0.324	0.172	0.593	0.473	0.900	0.453	0.393	

(0.500–0.860) indicate strong explanatory power, while Q^2 values (0.400–0.670) exceed the minimum threshold of 0.00, demonstrating the model's predictive relevance. The Standardized Root Mean Square Residual (SRMR) value of 0.078, being below the 0.08 threshold, signifies an acceptable model fit with minimal discrepancy between observed and predicted relationships. These quality assessment metrics validate the PLS-SEM approach as the optimal method for this study, providing robust, predictive insights into the infrastructural and technological determinants of OCM and OPD in SSA slums.

4.6. Robustness checks

The study used the SmartPLS4 software and applied the partial least squares structural equation modeling (PLS-SEM) method for data modeling. The data analysis process began with robustness checks to address any data issues that could affect the accuracy of the path coefficient estimates. Before this step, the data was tested for common method bias and nonresponse bias. Next, the study examined potential issues like endogeneity, quadratic effects, and unobserved heterogeneity

using the PLS approach. All these procedures are post-quality control measures to ensure data robustness and suitability for further analysis. Once the robustness checks were completed, the validity and reliability of the data were assessed by evaluating both the measurement and structural models (Ringle et al., 2018; Wong, 2013). This ensured the data was suitable for path analysis, which was used to test the relevance or irrelevance of the study's hypotheses. Additionally, the study followed the guidelines set by Hair et al. (2014) to confirm the data was adequately represented for PLS-SEM analysis.

4.6.1. Common method bias and nonresponse bias

The study used a collinearity test based on Kock (2015) criteria to check for common method bias (CMB) in the data. According to Kock (2015), if the inner Variance Inflation Factor (VIF) values are 3.3 or below, it indicates the absence of CMB. In this study, the inner VIF values ranged from 1.000 to 2.333, which are all below 3.3, confirming there was no CMB in the dataset. The study also examined the potential for nonresponse bias (NRB) during data collection to ensure that the respondents accurately reflected the characteristics of the entire population. To achieve this, purposive sampling was used, specifically targeting slum areas in SSA to make the findings more applicable to slum residents. While the direct approach to gathering responses helped improve the overall response rate, we carried out a multigroup analysis (MGA) to statistically test for the presence of NRB. In this analysis, respondents were divided into two groups: "early" respondents (those who completed the survey within the first two months, totaling 4,547) and "latter" respondents (those who completed the survey after two months, totaling 1,283). The goal was to see if any significant differences between these two groups could indicate bias. The empirical results from the MGA revealed no significant differences between the early and later respondents. This suggests that NRB would not affect the findings, confirming that the data collected from both groups accurately represented the overall population in the slum areas of SSA. By using this method, the study successfully minimized any potential biases and ensured that the insights gained were representative of the slum communities in SSA. The rigorous approach to detecting NRB strengthens the validity of the findings, making them more reliable for broader application to similar populations.

4.6.2. Endogeneity assessment

To examine potential endogeneity issues, this study applied the Gaussian copula (GC) method following Becker et al. (2022). Through bootstrapping, the analysis detected significant endogeneity effects between several constructs, with p-values below 0.05, indicating potential bias in these relationships. Specifically, endogeneity was found in the following pathways: STC→OCM ($\beta = -0.073$, $t = 2.212$, $p = 0.027$), STC→OPD ($\beta = -0.060$, $t = 2.459$, $p = 0.014$), DTD→OCM ($\beta = -0.100$, $t = 5.240$, $p = 0.000$), OPD→OCM ($\beta = -0.046$, $t = 2.300$, $p = 0.022$), and ADS→OCM ($\beta = 0.170$, $t = 8.310$, $p = 0.000$). These results indicate that the relationships between these constructs may be confounded by omitted variables or reciprocal causation, making it challenging to interpret causal effects accurately. To control for these endogeneity issues, the study generated instrumental variables (IVs) for the affected constructs—STC, DTD, OPD, and ADS—using principal component analysis (PCA) as recommended by Kock (2022). The resulting IVs, labeled STC_IV, DTD_IV, OPD_IV, and ADS_IV in Table 2 (Appendix), were designed to be exogenous, enabling them to effectively serve as controls for endogeneity in each respective construct. These IVs were then linked back to their corresponding constructs to allow for an indirect control mechanism. The GC bootstrapping approach was used to validate the IV effectiveness. The assessment revealed significant relationships between the constructs, suggesting that the IVs mitigated endogeneity appropriately. The relationships between constructs after including the GC instrumentals were: GC(DTD)→OCM ($\beta = -0.094$, $t = 5.266$, $p = 0.000$), GC(DTD)→OPD ($\beta = -0.031$, $t = 2.639$, $p = 0.008$), GC(ADS)→OCM ($\beta = 0.163$, $t = 8.284$, $p = 0.000$), GC(STC)→OCM ($\beta = -0.081$, $t = 2.669$, $p = 0.008$), GC(STC)→OPD ($\beta = -0.076$, $t = 5.956$, $p = 0.000$), and GC(OPD)→OCM ($\beta = -0.042$, $t = 2.140$, $p = 0.032$). The effectiveness of the IVs was further confirmed by strong relationships between the GC-adjusted IVs and their respective constructs, with values as follows: GC(STC_IV)→STC ($\beta = -0.009$, $t = 3.993$, $p = 0.000$), GC(ADS_IV)→ADS ($\beta = 0.001$, $t = 2.564$, $p = 0.010$), GC(OPD_IV)→OPD ($\beta = 0.239$, $t = 26.827$, $p = 0.000$), and GC(DTD_IV)→DTD ($\beta = 0.001$, $t = 3.413$, $p = 0.001$). These consistent relationships confirm that the IVs are suitable proxies to control for endogeneity and improve the reliability of causal interpretations. By implementing IVs through GC, the study effectively mitigates endogeneity, thereby ensuring that the observed relationships reflect more accurate associations, free from potential biases. The statistical outcomes confirming the suitability of IVs are presented in Table 2 (Appendix).

4.6.3. Quadratic effect assessment

Nonlinearities in data can complicate modeling, especially when relationships between constructs are more complex than a simple linear trend. To explore this, the study tested the quadratic (curvilinear) effects using Partial Least Squares (PLS) with bootstrapping. Initially, the results revealed significant quadratic effects between certain relationships: DTD to OCM ($\beta = -0.035$, $t = 4.953$, $p = 0.000$), STC to OCM ($\beta = -0.046$, $t = 4.151$, $p = 0.000$), and ADS to OCM ($\beta = 0.051$, $t = 6.670$, $p = 0.000$), suggesting potential curvilinear effects. To account for these quadratic effects, latent variables were generated by squaring the items of the constructs and labeling them: DTDsq, STCsq, and ADSsq as shown in Table 2 (Appendix). These newly created latent variables were incorporated into the model. A second round of bootstrapping was performed to assess the impact of these new variables. However, the results showed no significant effects for all three: QE(DTDsq) to OCM ($\beta = 0.048$, $t = 1.045$, $p = 0.296$), QE(STCsq) to OCM ($\beta = -0.223$, $t = 1.329$, $p = 0.184$), and QE(ADS2) to OCM ($\beta = 0.039$, $t = 1.580$, $p = 0.114$). This outcome suggests that the relationships between the constructs (DTD, STC, and ADS) and OCM are linear rather than curvilinear. This further implies that the initial quadratic effects were not statistically substantial. Therefore, the study concludes that a linear model better explains the relationships, simplifying the interpretation and supporting the assumption of straightforward, linear dynamics between the constructs and the outcome variable (OCM). The findings, confirming the adequacy of the quadratic latent variables, are summarized in Table 2 (Appendix).

4.6.4. Unobserved heterogeneity assessment

To explore unobserved heterogeneity in Partial Least Squares (PLS) path models, the finite mixture (FIMIX) PLS approach was applied. The process began by estimating the maximum number of potential segments, based on the minimum sample size required for segment estimation, as proposed by (Sarstedt et al., 2017). Using an effect size of 0.15 and a power of 80 %, the analysis determined that a minimum of 699 participants was necessary, allowing the model to estimate up to six segments. The FIMIX-PLS procedure was run across models with one to six segments, and several model selection criteria were used to evaluate the results: Akaike's Information Criterion (AIC), AIC with a factor of 3 (AIC3), consistent AIC (CAIC), AIC with a factor of 4 (AIC4), Bayesian Information Criterion (BIC), minimum description length with factor 5 (MDL5), and the normed entropy statistic (EN). The output, as shown in Table 3 (Appendix), presented mixed findings. AIC3 supported a six-segment model, while CAIC and BIC leaned toward a four-segment solution. AIC4, however, suggested the presence of only two segments, and MDL5 pointed to six segments as well. The inconsistency among these criteria indicates that there is no clear agreement on the appropriate number of segments to use. For instance, while AIC3 and MDL5 favored more segments, BIC and CAIC proposed fewer, leading to conflicting recommendations across the board. This variation in model selection criteria suggests that the presence of unobserved heterogeneity in the data may not be substantial enough to justify segmenting the

sample. After further robustness testing, it was concluded that the model's reliability is unaffected by the segmentation, and thus, the data will be treated as a single group for the final analysis. The empirical values of unobserved heterogeneity model criteria are shown in Table 3 (Appendix).

5. Research findings

5.1. Measurement model assessment and interpretation

This study leverages Structural Functionalism (SF) to synthesize findings, offering a holistic understanding of how infrastructure and technological factors shape online commerce (OCM) and product delivery (OPD) within slum environments. The study highlights the complex interdependencies among constructs such as addressing systems (ADS), transportation infrastructure (TPI), unreliable telecommunication (UTC), the digital divide (DTD), education and digital literacy (EDL), and security concerns (STC). These elements collectively form a systemic network that influences the accessibility and efficiency of OCM and OPD in slum economies. The descriptive statistics presented in Table 2 highlight the mean and standard deviation of the constructs, reflecting broad consensus among respondents regarding their significance. High mean values (above 3.8) suggest widespread recognition of their impact. For instance, ADS (3.811), TPI (4.028), EDL (3.882), and UTC (3.881) emerge as critical determinants of both OCM and OPD. The variability in responses, illustrated by higher standard deviations in constructs such as DTD (1.136) and ADS (0.992), suggests diverse experiences among respondents. This variation highlights unequal access to digital tools and internet connectivity, reflecting systemic infrastructural and technological disparities across different slums.

The measurement model as illustrated in Fig. 2 and Table 4, demonstrates the robustness and validity of the constructs. Factor loadings ($\lambda = 0.921–0.620$) predominantly exceed the 0.708 threshold, indicating strong item reliability. Although EDL1 ($\lambda = 0.620$) falls slightly below this benchmark, Hair et al. (2019) suggest that a minimum loading of 0.600 is acceptable in exploratory research, justifying its inclusion. Internal consistency is confirmed by high Cronbach's Alpha ($\propto = 0.881–0.768$), composite reliability (CR = 0.919–0.852), and Rho Alpha ($\rho A = 0.887–0.777$) values, all surpassing the 0.700 threshold. Convergent validity is established, with average variance extracted (AVE = 0.739–0.592) values exceeding the 0.5 benchmark, confirming that the constructs adequately capture their intended latent variables (Hair et al., 2019; Shiau et al., 2019). Additionally, low variance inflation factor (VIF = 3.669–1.372) values indicate minimal multicollinearity, ensuring that the constructs are statistically distinct (Hair et al., 2020; Kock and Lynn, 2012). These findings highlight the significant impact of infrastructural and technological barriers on slum e-commerce dynamics, suggesting that targeted interventions could enhance the efficiency and inclusiveness of OCM and OPD. Discriminant validity was further assessed using the Fornell-Larcker Criterion (FLC) and Heterotrait-Monotrait (HTMT) ratio (Henseler et al., 2015). The FLC results as presented in Table 3 show that the square root of AVE for each construct exceeds its correlations with other constructs, confirming their distinctiveness (Hair et al., 2019). Similarly, HTMT values in Table 4 remain below the 0.90 threshold, minimizing the risk of construct overlap (Hair et al., 2020). For example, ADS, DTD, and EDL exhibit clear boundaries with OCM and OPD, demonstrating that while these constructs influence OCM and OPD, they measure fundamentally different aspects. These results validate the robustness of the constructs and emphasize the unique role of infrastructure and technological factors in shaping slum e-commerce ecosystems. Collectively, these results confirm the robustness of the measurement model, highlighting the distinct and statistically sound influence of infrastructural and technological factors on online commerce (OCM) and online product delivery (OPD) in slum settings.

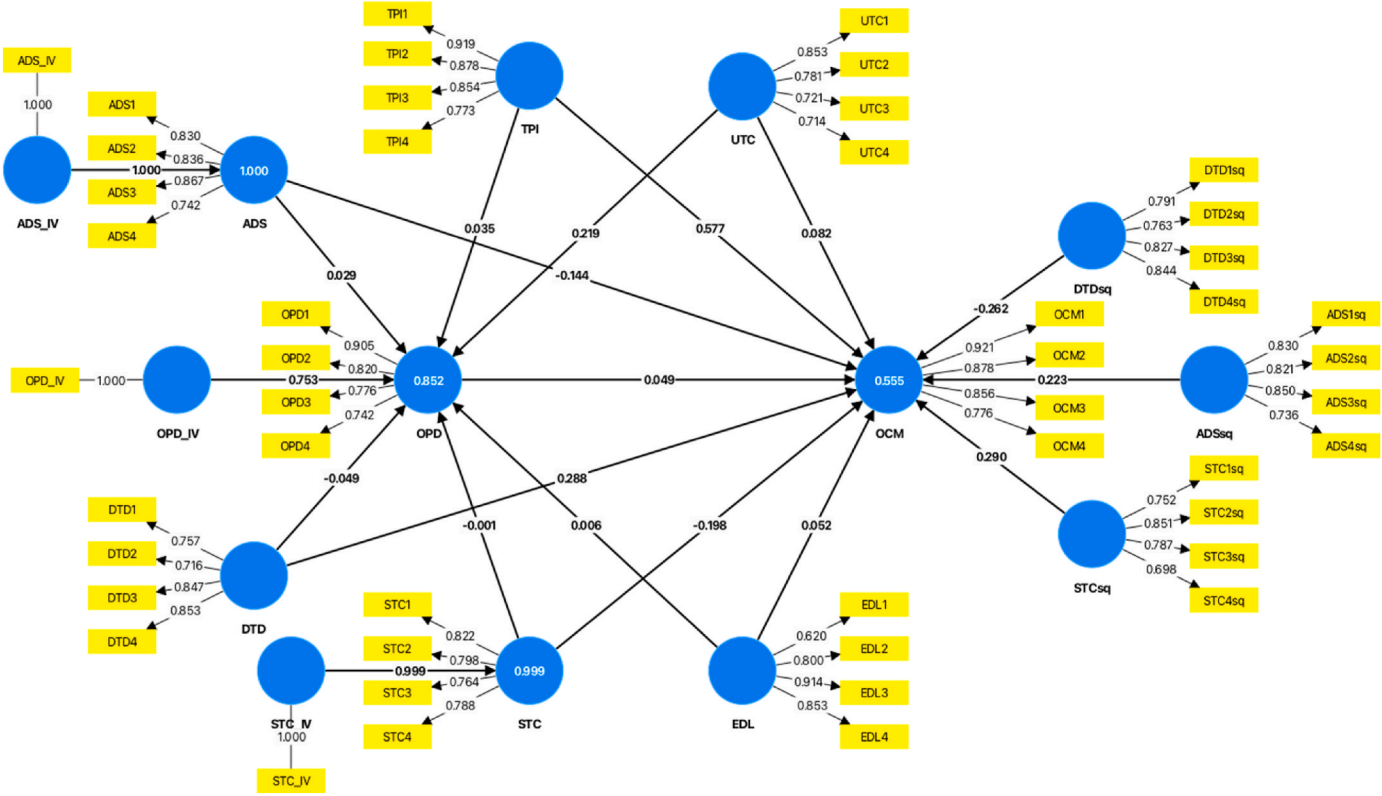


Fig. 2. The measurement model outcomes of impacting factors on slum OCM and OPD.

5.2. Structural model assessment and interpretation

The structural model assessment provides critical insights into the relationships among infrastructure, technological factors, online commerce (OCM), and online product delivery (OPD) within SSA slum communities. Per the recommendations of Hair et al. (2019), the model demonstrates substantial explanatory power, accounting for 55.5 % of the variance in OCM ($R^2 = 0.555$) and 85.2 % in OPD ($R^2 = 0.852$). This highlights the significant influence of constructs such as ADS, TPI, UTC, DTD, EDL and STC. The effect sizes (f^2) of the factors range from 0.551 (large effect) to 0.007 (minimal effect), revealing the varying degrees of influence among these factors (Hair et al., 2020; Sarstedt et al., 2017). For instance, constructs with large effect sizes (e.g., $f^2 > 0.35$) should be prioritized for targeted interventions, while those with minimal effect sizes (e.g., $f^2 < 0.15$) may require less immediate attention. The predictive relevance (Q^2) values for key constructs: 0.669 (ADS), 0.407 (OCM), 0.551 (OPD), and 0.626 (STC) confirm the model's reliability in anticipating outcomes (Geisser, 1974; Sarstedt et al., 2017; Shmueli and Koppius, 2011). All Q^2 values exceed the zero threshold, demonstrating meaningful predictive strength.

To ensure the robustness of the findings, bootstrapping with 5,000 resamples was applied in PLS-SEM path analysis, adjusting for skewness using Bias-Corrected and Accelerated (BCa) Bootstrap Confidence Intervals (95 %) (Aguirre-Urreta and Rönkkö, 2018). This approach confirms mediation when zero is excluded from the intervals, following the methodology of Carrión et al. (2017). Confidence intervals for the mediation effects are reported in Table 6, providing additional insights into the precision of the estimates. Path significance was determined using t-values (≥1.96), p-values (<0.05), and confidence intervals, with all direct, indirect, and total effects demonstrating statistical significance (Nitzl, 2016; Ringle and Sarstedt, 2016). The model's fit was further validated using the Standardized Root Mean Square Residual (SRMR), which yielded a value of 0.078, well below the recommended threshold of 0.08, indicating a good fit (Henseler et al., 2015). Together, these results confirm the robustness of the structural model, providing a strong foundation for interpreting the relationships among the constructs and emphasizing the systemic interdependencies that shape slum e-commerce ecosystems.

5.3. Hypothesis results

The study delved into the effects of infrastructural deficiencies and technological barriers within slum communities, examining how these factors influence online commerce (OCM) and product delivery (OPD) within these areas. Additionally, the research investigated the role of OPD as a mediator in the relationship between these factors and OCM.

Table 5
Results on the direct effects of infrastructure and technology on OCM and OPD.

	Relationship	β	SD	T statistics	P values	Decision
H1	OPD→OCM	0.048	0.012	4.128	0.000	Supported
H1a	ADS→OCM	−0.141	0.056	2.521	0.012	Supported
H1b	TPI→OCM	0.578	0.016	35.575	0.000	Supported
H1c	UTC→OCM	0.082	0.012	6.593	0.000	Supported
H1d	DTD→OCM	0.292	0.047	6.143	0.000	Supported
H1e	EDL→OCM	0.051	0.016	3.194	0.001	Supported
H1f	STC→OCM	−0.201	0.049	4.101	0.000	Supported
H2a	ADS→OPD	0.027	0.006	4.360	0.000	Supported
H2b	TPI→OPD	0.034	0.006	5.814	0.000	Supported
H2c	UTC→OPD	0.213	0.012	17.956	0.000	Supported
H2d	DTD→OPD	−0.047	0.007	6.795	0.000	Supported
H2e	EDL→OPD	0.006	0.008	0.811	0.417	Not Supported
H2f	STC→OPD	−0.001	0.007	0.187	0.852	Not supported

5.3.1. The effect of infrastructure and technology factors on OCM and OPD

Table 5 and Fig. 3 show the outcomes of the direct effects of infrastructural and technological barriers on OCM and OPD. From a structural-functionalism perspective, the significant positive relationship between OPD and OCM highlights OPD's role as a functional component, where its efficiency strengthens the broader e-commerce system. Infrastructure deficiencies, such as poor addressing systems (ADS), inadequate transportation (TPI), and unreliable telecommunication (UTC), act as systemic dysfunctions that negatively impact both OPD and OCM. These shortcomings reflect the structural limitations in slum environments, where basic infrastructural systems are either absent or fragmented, preventing optimal functioning of the e-commerce ecosystem. Technological factors show a nuanced impact: while the digital divide (DTD) significantly influences both OCM and OPD, issues like education and digital literacy (EDL) and security concerns (STC) do not exert meaningful influence on OPD. This suggests that while technology adoption barriers hinder the overall system's equilibrium, certain components like digital divides are more critical for functionality than others. The findings as shown in Table 5 and Fig. 3 emphasize the importance of prioritizing infrastructure improvements to enable OPD efficiency, which in turn supports OCM, while also addressing key technological enablers like DTD. By identifying critical dysfunctions and prioritizing targeted interventions, this analysis aligns with the structural functionalism lens, advocating for systemic enhancements to achieve an integrated and sustainable e-commerce framework within slum environments.

5.3.2. Mediation analysis

The mediation analysis, framed within structural functionalism, underscores Online Product Delivery (OPD) as a crucial mitigating mechanism that reduces the disruptive impact of infrastructural deficiencies on online commerce (OCM) in SSA slums. The results in Table 6 reveal that OPD plays a stabilizing role by partially absorbing the negative effects of inadequate addressing systems (ADS), transportation infrastructure (TPI), and unreliable telecommunications (UTC) while fully mediating the impact of security concerns (STC) on OCM. These findings highlight OPD's systemic function in integrating slum economies into digital commerce despite persistent infrastructural barriers. The analysis confirms partial mediation in the relationships between ADS, TPI, and UTC on OCM, indicating that while OPD mitigates these constraints, some residual inefficiencies remain. The mediation effect of ADS → OPD → OCM ($\beta = 0.001$, $p = 0.005$, CI [0.001, 0.002]) suggests that OPD reduces the challenges posed by inadequate addressing systems but does not fully eliminate them, meaning that location accuracy issues still impact e-commerce operations. Similarly, the mediation effect of TPI → OPD → OCM ($\beta = 0.002$, $p = 0.001$, CI [0.001, 0.003]) indicates that OPD improves last-mile logistics by compensating for weak transportation infrastructure, though some direct effects persist, emphasizing the need for structural improvements in road networks to optimize digital commerce participation. The strongest mediation effect is observed in UTC → OPD → OCM ($\beta = 0.010$, $p = 0.000$, CI [0.006, 0.016]), demonstrating that OPD significantly reduces the impact of unreliable telecommunications on digital transactions. However, since OPD cannot entirely eliminate the consequences of poor connectivity, stable telecommunications infrastructure remains essential for the full realization of e-commerce potential in slum

Table 6
The mediating role of OPD.

	Relationship	β	SD	T statistics	P values	mediation
H3a	ADS→OPD→OCM	0.001	0.003	2.815	0.005	Partial
H3b	TPI→OPD→OCM	0.002	0.001	3.228	0.001	partial
H3c	UTC→OPD→OCM	0.010	0.003	4.014	0.000	Partial
H3d	STC→OPD→OCM	0.000	0.000	0.182	0.856	Full

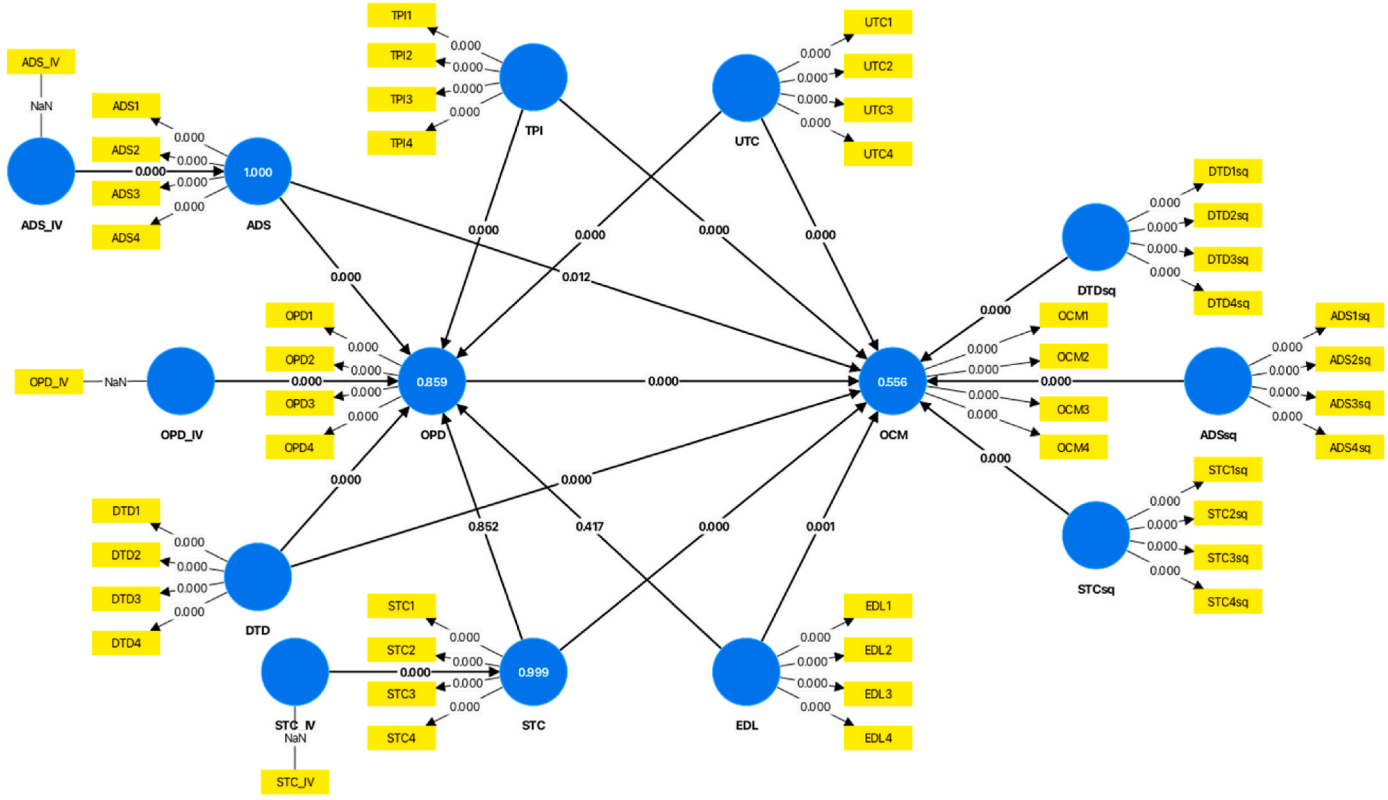


Fig. 3. The path analysis of the structural model of impacting factors on OPD and OCM.

environments. Unlike the other variables, STC → OPD → OCM (β = 0.000, p = 0.856, CI [-0.001, 0.001]) exhibits full mediation, meaning that security concerns alone do not directly impact OCM unless OPD is integrated as a trust-enhancing mechanism. This finding suggests that while infrastructural barriers create direct disruptions in digital commerce, security concerns function differently, requiring OPD as an intermediary to build trust and facilitate transaction reliability. Consequently, this highlights the necessity of additional systemic interventions such as improved cybersecurity measures, fraud prevention policies, and consumer protection frameworks to establish confidence in digital commerce within slum communities. The statistical significance of the mediation findings is also confirmed in Fig. 3, which presents the p-values and structural paths of the constructs.

6. Discussion

6.1. Effect of infrastructure and technological factors on slum OCM

The strong positive relationship between online product delivery (OPD) and online commerce (OCM), as supported by Hypothesis H1, highlights OPD as a pivotal enabler of digital commerce participation in slum communities. Structural Functionalism, as articulated by Parsons and Mayhew (1982), underscores that societal stability depends on the integration of functional subsystems, each contributing to systemic equilibrium. OPD, in this context, serves as a logistical driver that facilitates economic exchange, ensuring that marginalized populations can effectively engage in e-commerce. Its role transcends mere transportation, acting as a structural adaptation to infrastructural deficiencies such as unreliable addressing systems. This interdependence aligns with Merton (1957) differentiation between manifest and latent functions. The manifest function of OPD is to transport goods efficiently, enhancing consumer satisfaction and market reliability. However, its latent function extends further, promoting digital inclusion and economic participation among slum residents. By overcoming barriers to access, OPD strengthens the structural cohesion of e-commerce, allowing informal economies to integrate into formal digital marketplaces. This reinforces Giddens' structuration theory, which extends Structural Functionalism by emphasizing the dynamic interplay between agency and structure in shaping economic behaviors (Jones and Karsten, 2008). The empirical findings resonate with Ignat and Chankov (2020), who argue that last-mile delivery determines e-commerce viability by influencing consumer trust and transaction completion rates. Similarly, Ali and Bhasin (2019) emphasize the mediating role of delivery efficiency in fostering consumer loyalty. However, contrasting perspectives from Surjandy et al. (2021) suggest that OPD's influence on purchasing intent can be disrupted by external crises, reflecting Merton (1957) notion of unanticipated consequences within social systems. Ultimately, OPD functions as a stabilizing force in the digital economy, but its effectiveness hinges on structural adaptability. Ensuring resilience against external disruptions such as economic crises or technological failures remains crucial for sustaining equitable access to e-commerce in marginalized settings.

The inadequacy of addressing systems (ADS) in slum communities presents a fundamental challenge to the efficiency of online product delivery (OPD), as evidenced by Hypothesis H1a. Unlike conventional logistical barriers, ADS deficiencies reflect a deeper systemic issue within urban infrastructure, disrupting the interconnectivity necessary for digital commerce. Structural Functionalism, particularly Parsons' concept of system integration, emphasizes that functional subsystems such as infrastructure must operate cohesively for economic transactions to thrive (Ormerod, 2020). When addressing systems are underdeveloped, delivery networks become fragmented, leading to inefficiencies that undermine the growth of online commerce. Merton (1957) distinction between manifest and latent dysfunctions further explains how the absence of structured ADS impacts e-commerce. The manifest dysfunction is the failure to locate delivery destinations, leading to lost shipments and delayed transactions. Latently, this deficiency discourages businesses from investing in slum-based delivery networks,

reinforcing digital exclusion. Empirical studies by Steinfield and Whitten (2006) highlight that unreliable ADS not only increases operational costs but also deters e-commerce expansion in underserved regions. Giddens' structuration theory extends this analysis by illustrating how infrastructure constraints shape consumer behavior (Jones and Karsten, 2008). Slum residents adapt by relying on informal drop-off points or community-based delivery hubs, but these workarounds do not resolve the systemic dysfunction. To restore equilibrium, interventions such as digital geolocation tools and community-led mapping initiatives are necessary. Strengthening ADS is not just about improving logistics but also enabling economic participation and fostering long-term digital inclusion for marginalized communities.

The findings of Hypothesis H1b show that the inefficiencies in transportation infrastructure (TPI) pose a major obstacle to the expansion of online commerce (OCM) in slum communities, where logistical challenges often outweigh the benefits of digital transactions. Unlike in well-developed urban areas where structured transport systems support seamless delivery, slum environments suffer from inadequate roads, unreliable public transport, and inefficient last-mile logistics. Structural Functionalism provides a lens to understand these disruptions. Parsons and Mayhew (1982) argue that societal subsystems must function cohesively to maintain stability. However, when transportation fails, the entire e-commerce ecosystem is destabilized, making product delivery unreliable and costly. Beyond delivery delays, inadequate TPI leads to high transportation costs, failed shipments, and logistical inefficiencies, ultimately diminishing consumer trust and engagement with online markets. Merton (1957) theory of dysfunction explains how structural failures reinforce economic exclusion, as unreliable delivery systems make digital commerce inaccessible for slum residents. Supporting studies highlight these systemic impacts. Qi et al. (2020) found that weak transportation infrastructure negatively affects supply chain reliability, while Cedillo-Campos et al. (2022) emphasize how fragmented logistics systems hinder last-mile delivery in underserved areas. Similarly, Mesjasz-Lech and Włodarczyk (2022) demonstrate that transportation inefficiencies significantly limit e-commerce adoption in informal settlements. While some businesses attempt workarounds such as using boda-boda couriers, community pick-up hubs, or informal delivery networks, these adaptations are unsustainable and fail to address root structural deficiencies. Sustainable solutions require investment in road networks, hybrid logistics models integrating formal and informal delivery systems, and urban planning policies tailored to slum environments. Without these interventions, the full potential of OCM in slum communities will remain unrealized, perpetuating digital and economic marginalization.

The impact of unreliable telecommunications (UTC) on online commerce (OCM) (H1c) in slum communities reveals a deeper structural imbalance that Structural Functionalism helps to illuminate. Telecommunications, as a critical subsystem, is expected to facilitate market access, digital transactions, and economic inclusion. However, its failure disrupts the interconnected functions of e-commerce, reinforcing digital exclusion and economic marginalization. Unlike conventional economic models that view telecommunications simply as an enabler of commerce, Structural Functionalism positions it as a stabilizing force necessary for systemic equilibrium (Izadi et al., 2020). The absence of reliable ICT infrastructure in slums reflects a broader dysfunction, where institutional neglect worsens socio-economic disparities rather than alleviating them. Merton's concept of dysfunction is particularly relevant here. Thus, rather than serving its intended function of connectivity, UTC becomes a structural barrier, limiting residents' ability to engage with online markets (Crothers, 2004). The study's findings corroborate the outcomes of Xing (2018) and Jehangir et al. (2011), which highlight how ICT failures in marginalized areas directly correlate with lower e-commerce adoption and economic stagnation. Moreover, research by Liu and Yang (2010) emphasizes that access to stable telecommunications is a key determinant of digital commerce success, reinforcing the argument that slums face systemic barriers beyond individual technological literacy. While informal solutions such as offline ordering methods and community networks, temporarily mitigate these limitations, they do not resolve the underlying structural deficiency. A functionalist approach suggests systemic interventions through investment in ICT infrastructure, inclusive digital policies, and adaptive technological solutions to restore equilibrium, ensuring slum populations can meaningfully participate in the digital economy.

Hypothesis H1d underscores the critical influence of disparities in digital technology access and usage on the development of online commerce (OCM) in slum areas. Digital exclusion in slum communities represents more than just a technological gap. It signifies a deeper structural dysfunction within the socio-economic system. Structural Functionalism provides a useful framework for understanding this imbalance, as it posits that society functions through interdependent subsystems that must work cohesively to maintain stability (Ormerod, 2020; Parsons and Mayhew, 1982). When digital infrastructure fails to integrate marginalized populations, it disrupts this equilibrium, reinforcing patterns of economic and social exclusion. Merton (1957) concept of latent dysfunctions further explains how technological advancements, while intended to enhance accessibility, can inadvertently widen disparities when slum communities are left behind. In SSA, where unreliable internet access and low smartphone penetration persist, these systemic failures prevent e-commerce from serving as an inclusive economic force. Sánchez-Torres (2019) and Murthy et al. (2021) highlight how digital exclusion restricts consumer engagement, limiting both market participation and economic mobility. Beyond economic barriers, digital inequalities also disrupt broader societal functions. Smelser (2005) theory of structural differentiation suggests that as societies evolve, specialized subsystems (e.g., digital commerce) must be accessible to all for equilibrium to persist. In slum environments, however, the digital divide exacerbates existing inequalities, creating a fragmented economic landscape. To restore balance, systemic interventions such as expanding affordable technology, enhancing digital education, and investing in infrastructure are necessary. By addressing these structural deficiencies, OCM can transition from an exclusive privilege to an inclusive economic stabilizer, fostering greater participation and reducing socioeconomic disparities in marginalized communities.

The significance of Hypothesis H1e concerning online commerce (OCM) in slum communities underscores the crucial role of education and digital literacy (EDL) as a foundational element in the socioeconomic structure. Education and digital literacy (EDL) are more than just personal assets; they are structural prerequisites for economic participation in the digital age. Within the Structural Functionalist paradigm, education serves as a key integrative subsystem that connects individuals to the broader socio-economic framework (Parsons and Mayhew, 1982). However, in slum communities, systemic inadequacies in EDL disrupt this equilibrium, preventing residents from effectively engaging in online commerce (OCM). The absence of digital literacy creates functional disconnections between individuals and the economic structures that rely on technological engagement, leading to a cycle of exclusion. Merton (1957) concept of dysfunction provides a lens to analyze how the failure to integrate digital literacy into slum communities perpetuates economic marginalization. Research by Valarezo et al. (2020) and Mahmood et al. (2022) reinforces this by showing that individuals with low digital literacy are significantly less likely to engage in e-commerce. Furthermore, Smelser (2005) theory of structural differentiation highlights that as economic systems become increasingly digital, exclusion from this transformation exacerbates inequality. To address this systemic dysfunction, targeted interventions must focus on reintegrating EDL as a stabilizing mechanism. Strategies could include embedding digital literacy into local curricula, establishing community learning hubs, and providing accessible training programs on e-commerce tools. These measures would not only enable slum residents to participate in the digital economy but also strengthen the functional integration of OCM as a transformative force for economic inclusion.

The findings of Hypothesis H1f reveal the profound impact of

security concerns (STC) on online commerce (OCM) in slum communities. Concerns over security in online commerce (STC) in slum communities reflect more than just technological shortcomings; they signify a systemic breakdown in the trust mechanisms necessary for economic participation. Structural Functionalism, as articulated by Ormerod (2020) and Kingsbury and Scanzoni (2009), asserts that a well-functioning society depends on the stability and integration of its subsystems. However, in slum environments, the absence of reliable consumer protection, weak regulatory enforcement, and distrust in digital financial systems disrupt this equilibrium, leading to disengagement from online commerce. Merton (1957) concept of dysfunction is evident here—while digital platforms aim to democratize access to commerce, inadequate security infrastructure creates unintended barriers. Residents, perceiving high risks of fraud, financial loss, and data misuse, rationally avoid online transactions. This aligns with Anic et al. (2019) findings that insufficient consumer protection fosters digital mistrust. Furthermore, Jibril et al. (2020) demonstrate how fears of financial loss (FOFL) and reputational damage (FORD) disproportionately affect economically vulnerable populations. Smelser (2005) theory of social differentiation further contextualizes this problem. As digital economies evolve, slum communities remain systematically excluded due to unresolved structural deficiencies. Addressing these dysfunctions requires systemic recalibration. Enhancing cybersecurity policies, integrating digital fraud awareness into local education systems, and tailoring consumer protection frameworks to marginalized populations can restore equilibrium. Without these interventions, slum residents will continue to be sidelined in the digital economy, reinforcing systemic inequalities rather than bridging them.

6.2. Effect of infrastructure and technological factors on slum OPD

The relationship between inadequate addressing systems (ADS) and online product delivery (OPD) in slum communities, as evidenced by Hypothesis H2a, highlights a critical systemic dysfunction from a Structural Functionalism perspective. According to Mishra and Rath (2020), Durkheim's concept of organic solidarity posits that social systems rely on interdependent structures for stability. The inefficiency of ADS disrupts the equilibrium of the e-commerce ecosystem, creating logistical barriers that hinder digital market participation. Parsons' AGIL framework according to Ormerod (2020) further explains this dysfunction, as ADS fails to fulfill its adaptation function, preventing seamless interaction between OPD and OCM. This is consistent with studies by Xiao et al. (2018) and Nabot and Garaj (2014), which demonstrate that poor addressing systems increase delivery inefficiencies, eroding consumer trust. Merton (1957) notion of dysfunctions applies to the way ADS deficiencies create unintended negative consequences for OCM, restricting economic integration in slum communities. While OPD functions as an adaptive subsystem, it remains constrained by the limitations imposed by ADS, leading to inefficiencies that ripple through the broader e-commerce system. These systemic weaknesses reinforce the need for institutional reforms that strengthen addressing infrastructures rather than relying solely on OPD adaptations. Ultimately, the findings highlight the necessity of a holistic intervention that integrates logistical innovations with infrastructural reforms. Improving ADS would enhance OPD efficiency, reinforcing the interdependence of subsystems within the e-commerce ecosystem. By addressing root causes rather than temporary adaptations, slum communities can achieve greater economic participation and digital inclusion, aligning with Structural Functionalism's emphasis on maintaining systemic stability through institutional development.

The findings from Hypothesis H2b underscore the structural dysfunction created by inadequate transportation infrastructure (TPI) in slum communities, revealing its profound impact on online product delivery (OPD) and the broader e-commerce ecosystem. In the context of Structural Functionalism, as highlighted by Kingsbury and Scanzoni (2009), TPI functions as a critical structural component that enables efficient logistical operations. However, in slum environments, the absence of paved roads, unreliable transport networks, and congestion disrupts the equilibrium necessary for seamless product movement, leading to inefficiencies such as delivery delays, increased costs, and reduced consumer trust in digital commerce. From a different perspective, these infrastructural deficiencies represent a latent dysfunction (Merton, 1957; Sohlberg, 2017), wherein the inability of TPI to support OPD hinders the expected benefits of e-commerce. This dysfunction not only obstructs the operational efficiency of last-mile logistics but also reinforces economic exclusion by limiting residents' access to digital marketplaces. Durkheim's theory of organic solidarity (Mishra and Rath, 2020) highlights that advanced economies thrive on the interdependence of subsystems. However, when TPI is deficient, it weakens this interconnectedness, preventing slum-based consumers and businesses from fully integrating into the broader digital economy. Empirical studies corroborate these findings. Banerjee et al. (2020) assert that improved transport systems bridge economic disparities by linking marginalized areas to formal supply chains. Similarly, Wang et al. (2020) emphasize that robust infrastructure enhances service accessibility and commercial participation. However, in slum settings, OPD alone cannot rectify TPI inadequacies. Instead, systemic interventions such as localized infrastructure investments, regulatory support for informal delivery networks, and adaptive urban planning are essential for strengthening OPD's functionality. By addressing these structural inefficiencies, policymakers can foster a more integrated and resilient e-commerce ecosystem, ensuring equitable access to digital marketplaces for underserved populations.

The relationship between unstable telecommunications infrastructure (UTC) and online product delivery (OPD) in slum communities, as demonstrated by Hypothesis H2c, highlights a systemic dysfunction from a Structural Functionalism perspective. Drawing on the foundational work of Parsons and Mayhew (1982), telecommunications infrastructure serves as a vital subsystem that enables communication, coordination, and tracking within e-commerce ecosystems. According to Ormerod (2020), Parsons' concept of functional prerequisites emphasizes the necessity of stable UTC for the survival and efficiency of digital marketplaces. However, in slum environments, frequent network disruptions and slow speeds hinder OPD efficiency, leading to delays, tracking failures, and reduced consumer trust. These disruptions exemplify Merton's concept of strain, where structural deficiencies prevent the system from achieving its intended functions (Crothers, 2004). The inability to maintain stable telecommunications weakens socio-economic cohesion, limiting slum residents' participation in digital commerce and reinforcing economic exclusion (Bernburg, 2019). This aligns with Merton's theory of anomie, where systemic inefficiencies create barriers to upward mobility. The empirical findings align with the work of Shamsuzzoha et al. (2016), who demonstrated that reliable ICT solutions enhance delivery efficiency and innovation by fostering better communication among stakeholders. Similarly, Ismail and Jokonya (2023) highlighted the critical role of stable e-payment and traceability systems in improving e-logistics satisfaction. In slum contexts, the absence of stable telecommunications perpetuates digital exclusion, widening disparities between affluent and underserved populations. To address this dysfunction, Structural Functionalism suggests systemic interventions that restore equilibrium. Parsons' emphasis on adaptive upgrading—the process by which systems evolve to meet new challenges—provides a theoretical foundation for interventions such as deploying low-cost, high-reliability network solutions tailored to the unique challenges of slum regions. Enhancing telecommunications infrastructure would reinforce OPD's role as a stabilizing mechanism, ensuring greater efficiency, consumer trust, and socio-economic integration in digital commerce.

The digital divide (DTD) in slum communities, as highlighted in Hypothesis H2d, represents a fundamental barrier to the integration of online product delivery (OPD) into everyday commerce. Unlike conventional infrastructural challenges, which often have tangible and

direct solutions, the digital divide is an evolving systemic dysfunction that permeates multiple layers of economic and social organization. From a Structural Functionalist perspective, Parsons and Mayhew (1982) argue that technological access functions as a prerequisite for societal adaptation and progression, shaping interactions between economic actors. In slum settings, however, limited access to smartphones, stable internet, and digital literacy disrupts this adaptive function, creating inefficiencies in OPD that hinder trust and participation in e-commerce. Merton's distinction between manifest and latent functions further explains how the digital divide operates in slum economies (Sohlberg, 2017). While the manifest function of digital technology is to facilitate efficient commercial exchanges, its latent dysfunction which is exclusion due to uneven digital access emerges as an unintended consequence that exacerbates economic disparities (Crothers, 2004). This limitation means that even when OPD systems exist, they remain underutilized by digitally disconnected consumers, leading to fragmented adoption patterns and reinforcing economic marginalization. Unlike traditional infrastructure gaps, which can be bridged with physical interventions, the digital divide demands a behavioral and institutional response, aligning with Parsons' notion of pattern maintenance. A concept that emphasizes the need for societies to sustain functional norms and values (Munch, 1982). The study's finding supports O'Sullivan and Walker (2018), who argue that unequal digital access restricts not only participation in online transactions but also broader economic integration, further deepening marginalization in underprivileged communities. Similarly, Ismail and Jokonya (2023) emphasize that digital inequalities directly impact e-logistics performance, limiting consumers' ability to track deliveries and engage in seamless e-commerce experiences. Addressing this dysfunction requires a more integrated approach where interventions extend beyond infrastructure provision to include socialization efforts that embed digital engagement into local economic behaviors. This involves targeted digital literacy initiatives, incentives for smartphone adoption, and OPD models that cater to non-digital users, ensuring that slum communities are not passively shaped by technological advancements but actively incorporated into them.

The statistical insignificance of Hypothesis H2e, examining the relationship between education and digital literacy (EDL) and online product delivery (OPD), challenges prevailing assumptions about digital inclusion in e-commerce ecosystems. Structural Functionalism, as outlined by Parsons and Mayhew (1982), posits that institutional subsystems such as education and infrastructure must operate cohesively for systemic stability. However, the findings suggest that while EDL is important, its influence on OPD is overshadowed by more pressing infrastructural deficiencies, such as inadequate addressing systems (ADS) and poor transportation infrastructure (TPI). This indicates that digital literacy alone cannot compensate for systemic dysfunctions, a notion aligned with Merton's concept of functional prerequisites, where subsystems must work in unison for optimal functionality. Contrary to previous studies emphasizing digital literacy as a driver of e-commerce participation (Bonina et al., 2021; Hussain, 2016), the results highlight the emergence of latent functions (Merton, 1957; Sohlberg, 2017). Informal logistics networks, reliance on intermediaries, and alternative ordering methods (e.g., phone calls and SMS) serve as compensatory mechanisms, ensuring OPD continuity despite digital literacy gaps. This reflects Alexander (2017) theory of adaptive structural responses, where informal institutions emerge to fill systemic voids. Furthermore, the findings align with Giddens' (1984) structuration theory, which extends Structural Functionalism by illustrating the recursive relationship between structure and agency (Jones and Karsten, 2008). In slum economies, structural barriers shape digital behaviors, limiting the effectiveness of individual digital literacy. Businesses adapt by prioritizing logistical workarounds over digital capacity-building, reinforcing the dominance of macro-level constraints over micro-level advancements. Thus, while EDL is essential for long-term digital inclusion, its immediate impact on OPD remains limited by structural inefficiencies. Addressing foundational infrastructural barriers is imperative before digital literacy initiatives can significantly enhance e-commerce participation.

The statistical insignificance of Hypothesis H2f, which examines the relationship between security concerns (STC) and online product delivery (OPD), provides a nuanced perspective on systemic priorities in slum-based e-commerce. Structural Functionalism, as articulated by Parsons and Mayhew (1982), posits that security operates as a regulatory subsystem essential for maintaining order. However, the findings indicate that STC does not significantly disrupt OPD outcomes, challenging traditional assumptions about the centrality of security in e-commerce logistics. A plausible explanation lies in the hierarchy of systemic dysfunctions. In slum environments, where logistical barriers such as inadequate addressing systems (ADS), unreliable telecommunications (UTC), and poor transportation infrastructure (TPI) dominate, security concerns may be deprioritized. As Merton (1957) suggests in his discussion of manifest and latent functions, the overt function of security in enabling trust and preventing fraud may be overshadowed by latent dysfunctions related to infrastructural inadequacies, which create more immediate delivery challenges. Consumers and businesses may prioritize ensuring product arrival over mitigating security risks, rendering STC's impact less pronounced. Additionally, businesses in high-risk environments often develop adaptive mechanisms that minimize the direct effect of STC. This aligns with Alexander (2017) theory of adaptive structural responses, which posits that informal adjustments emerge to counteract systemic inefficiencies. Strategies such as cash-on-delivery, designated collection points, and reliance on local intermediaries compensate for security gaps, ensuring OPD continuity despite underlying risks. These findings contrast with existing literature emphasizing security as a cornerstone of e-commerce success (Leon et al., 2023; Zhou et al., 2020). However, in slum contexts, security's relevance may be masked by more pressing infrastructural dysfunctions. This suggests a sequential dependency, where only after foundational logistics are strengthened will security concerns become a dominant factor in shaping OPD efficiency. Addressing broader systemic constraints may, therefore, be a prerequisite for STC to exert a measurable influence.

6.3. *The mediating role of OPD as mitigator*

While online product delivery (OPD) is widely recognized for its role in reducing lead time and enhancing customer satisfaction, its capacity to mediate slum-specific infrastructural constraints has remained underexplored in prior research. Existing studies on mediation effects in e-commerce primarily focus on variables such as customer satisfaction, trust, and perceived value (Agarwal et al., 2022; Ginting et al., 2023; Sitthipon et al., 2022; Zhai and Chen, 2023). This study introduces a novel perspective by examining OPD as a mitigating factor within slum e-commerce ecosystems, addressing the interplay between infrastructural challenges and online commerce (OCM). The findings reveal that infrastructural and technological constraints directly affect both OPD and OCM, with notable exceptions like EDL and STC on OPD. Despite these constraints, OPD demonstrates a positive influence on OCM, highlighting its partial mediation capacity in mitigating certain infrastructural effects. Conversely, full mediation outcomes suggest that some infrastructural issues impacting OCM exceed OPD's capabilities to resolve. These findings are contextualized within the structural functionalism framework, demonstrating OPD's critical yet bounded role in stabilizing slum e-commerce ecosystems.

The partial mediation of OPD in the relationship between ADS and OCM (H3a) underscores the adaptive role of OPD in maintaining the functionality of the e-commerce ecosystem amidst systemic dysfunctions. From a Structural Functionalist perspective, OPD acts as a stabilizing mechanism that compensates for the inefficiencies of inadequate addressing systems, ensuring that products reach customers despite infrastructural gaps. This aligns with Parsons and Mayhew (1982)

conceptualization of structural functionalism, where institutions evolve to maintain societal stability by addressing dysfunctions within subsystems. Similarly, Merton (1957) notion of manifest and latent functions illustrates how OPD explicitly facilitates commerce while implicitly mitigating structural inadequacies in addressing systems. However, the partial nature of this mediation reveals the limitations of relying solely on OPD innovations without addressing the foundational issues in slum addressing systems. This highlights the need for a dual approach: enhancing OPD strategies to improve delivery efficiency while implementing structural reforms to address the root causes of ADS inefficiencies. Such a synergistic strategy aligns with the Structural Functionalism principle of systemic integration, fostering a more inclusive and sustainable e-commerce ecosystem in slum environments.

The mediation effect of OPD in the relationship between TPI and OCM (H3b) further reinforces its functional role in sustaining e-commerce operations in slum environments (Parsons and Mayhew, 1982). posited that social systems maintain equilibrium through adaptive mechanisms that counteract systemic dysfunctions. In this context, OPD functions as an adaptive subsystem that mitigates logistical inefficiencies caused by poor TPI, ensuring the continuity of online commerce. Slum communities often face infrastructural bottlenecks, such as poor road networks, inadequate transportation services, and traffic congestion, which hinder product delivery efficiency. OPD addresses these dysfunctions through informal logistics networks, flexible delivery models, and localized distribution strategies, mirroring Sohlberg (2017) and Merton (1957) distinction between intended and unintended functional adaptations within complex systems. However, the persistence of structural limitations means OPD cannot fully absorb the inefficiencies of weak transportation infrastructure. This necessitates a complementary strategy that enhances OPD efficiency while pursuing infrastructural improvements to support long-term e-commerce growth. In line with Structural Functionalism's emphasis on systemic interdependence, integrating OPD with targeted infrastructural reforms will facilitate a more robust and sustainable digital commerce framework within slum economies.

The mediation (H3c) role of OPD in addressing the challenges posed by unstable telecommunications (UTC) infrastructure further illustrates its function as a stabilizing mechanism within the e-commerce ecosystem. Irfan (2023) contribution to the framework suggests that subsystems must adapt to external disruptions to maintain systemic stability. In the case of slum-based e-commerce, unreliable network connectivity disrupts real-time communication and transaction flows, posing significant barriers to OCM efficiency. OPD mitigates these disruptions by implementing alternative communication strategies, such as multi-channel coordination, predictive dispatching, and adaptive routing to ensure timely deliveries despite telecommunications setbacks. This aligns with Merton (1957) functional analysis, where OPD serves a manifest function of delivery while also fulfilling a latent role in sustaining consumer trust in digital commerce. However, given the partial mediation effect, OPD alone cannot fully offset the negative consequences of UTC inefficiencies. Addressing these structural challenges requires an integrated approach that enhances OPD resilience while investing in telecommunications infrastructure improvements. This dual approach aligns with Structural Functionalism's emphasis on systemic integration, ensuring a stable and adaptive digital commerce ecosystem within slum environments.

The mediation analysis for H3d indicates a full mediation effect, demonstrating that security concerns (STC) do not directly hinder online commerce (OCM) but exert their influence through online product delivery (OPD). This finding aligns with the Structural Functionalism perspective, which emphasizes the role of specialized subsystems in maintaining systemic stability (Parsons, 1951). In this case, OPD functions as an adaptive mechanism that counterbalances security-related dysfunctions, ensuring the continuity of OCM in slum environments where risks of theft, fraud, and delivery uncertainties are prevalent. According to Pope (1975) on Durkheim's structural functionalism, social structures rely on interdependent institutions to sustain equilibrium, with each component addressing specific systemic challenges. OPD exemplifies this principle by absorbing the security risks inherent in slum-based e-commerce and implementing structured solutions such as secure drop-off points, identity verification, and digital tracking systems. These mechanisms enhance consumer trust and facilitate transactional reliability, reinforcing the structural balance within the digital marketplace. By acting as an intermediary between STC and OCM, OPD stabilizes the e-commerce ecosystem, allowing it to function despite external threats. Merton (1957) concept of manifest and latent functions further illuminates OPD's role in mitigating security risks. While the manifest function of OPD is to facilitate the physical movement of goods, its latent function emerges as a trust-building tool that enhances consumer confidence in digital transactions. This unintended yet critical function addresses security apprehensions, ensuring that customers engage with e-commerce platforms despite prevailing risks. However, the full mediation effect also exposes the limitations of OPD as a standalone solution—while it mitigates immediate security concerns, it does not address deeper structural deficiencies such as weak law enforcement, inadequate fraud prevention frameworks, and insufficient consumer protection policies. To achieve systemic integration, Structural Functionalism suggests a coordinated approach where institutional reforms complement logistical innovations. Strengthening cybersecurity measures, enforcing stricter e-commerce regulations, and implementing fraud prevention technologies (e.g., blockchain for secure transactions) would reinforce OPD's stabilizing role. Such structural enhancements ensure that security risks do not become systemic dysfunctions that destabilize the e-commerce ecosystem. By harmonizing OPD's trust-enhancing functions with broader regulatory and technological advancements, policymakers and industry stakeholders can foster a resilient and inclusive digital economy in slum environments.

7. Conclusion

This study underscores the pivotal role of infrastructural and technological factors in shaping the dynamics of online commerce (OCM) and online product delivery (OPD) within slum environments. The findings reveal that infrastructural deficiencies, such as inadequate addressing systems (ADS), poor transportation infrastructure (TPI), and unreliable telecommunications (UTC), act as significant barriers, disrupting the efficiency of both OCM and OPD. Similarly, technological challenges, including the digital divide (DTD), education and digital literacy (EDL), and security concerns (STC), adversely impact OCM, though EDL and STC were found to have no significant direct effect on OPD. From a Structural Functionalist perspective, these challenges represent systemic dysfunctions that destabilize the e-commerce ecosystem. However, the study also highlights the adaptive role of OPD as a mediating mechanism, which partially mitigates the negative effects of infrastructural gaps, particularly in the relationships between ADS, TPI, UTC, and OCM. This demonstrates OPD's capacity to maintain system functionality despite structural deficiencies. Notably, security concerns (STC) present a unique scenario, where OPD serves as a full mediator, indicating that STC only impacts OCM when OPD is integrated as a trust-enhancing intermediary. This finding emphasizes the critical role of OPD in building transaction reliability and consumer confidence in environments plagued by security vulnerabilities. The study offers both practical and theoretical contributions. Practically, it calls for targeted interventions to address infrastructural and technological barriers, such as improving addressing systems, enhancing transportation networks, and bridging the digital divide. Theoretically, it advances the application of Structural Functionalism to e-commerce in marginalized contexts, illustrating how subsystems like OPD adapt to systemic dysfunctions to sustain ecosystem functionality. However, the findings also highlight the limitations of relying solely on logistical innovations like OPD, underscoring the need for complementary measures such as regulatory reforms, cybersecurity enhancements, and community-based

digital literacy programs to address root causes and foster a more inclusive and resilient e-commerce ecosystem in slum environments.

7.1. Practical implications

The study's findings reveal critical barriers to online commerce (OCM) and online product delivery (OPD) in slum communities, including inadequate addressing systems (ADS), poor transportation infrastructure (TPI), unreliable telecommunications infrastructure (UTC), and digital divide (DTD). To address these challenges, the study proposes detailed, actionable policy recommendations and business strategies, explicitly linked to the statistical findings, while expanding on emerging technologies and differentiating between short- and long-term solutions.

The significant negative impact of inadequate ADS on OCM (H1a) and OPD (H2a) underscores the need for scalable, low-cost digital addressing solutions. Policy Recommendation: Governments should launch a Slum Addressing Initiative (SAI), partnering with NGOs and tech companies to deploy GPS-based digital addressing systems. This initiative should begin with Phase 1: Community engagement and training to map informal settlements using mobile apps and low-cost GPS devices. Phase 2: Pilot testing in a specific slum area, with real-time feedback from residents and delivery agents. Phase 3: Scaling up based on pilot results, funded through public-private partnerships. Business Strategy: E-commerce platforms should integrate these digital addresses into their delivery systems, offering incentives such as discounts or loyalty points to customers who adopt the new addressing system. Emerging technologies like AI-driven geospatial mapping and blockchain-based verification can enhance precision and trust, ensuring long-term sustainability. In the short term, pilot projects can demonstrate feasibility, while long-term strategies focus on nationwide scaling and integration with national addressing systems.

The adverse effects of poor TPI on OCM (H1b) and OPD (H2b) highlight the need for innovative logistics solutions. Policy Recommendation: Governments should establish Localized Delivery Networks (LDNs), employing slum residents as micro-logistics agents equipped with lightweight delivery carts or electric scooters. These networks should be supported by modular logistics hubs powered by solar energy, strategically positioned at the periphery of slum areas to serve as aggregation points for last-mile delivery. Business Strategy: E-commerce companies should partner with LDNs, integrating mobile applications for real-time tracking and dynamic rerouting based on local data. Emerging technologies like route optimization algorithms and autonomous delivery drones can further improve efficiency, particularly in densely populated and unstructured slum environments. In the short term, pilot projects can test the feasibility of micro-logistics networks, while long-term strategies focus on scaling up and integrating electric vehicles and decentralized delivery models.

The study also identifies unreliable UTC as a significant barrier to OCM (H1c) and OPD (H2c). Policy Recommendation: Governments and telecommunications providers should collaborate to install affordable 4G or 5G infrastructure in slum areas, complemented by community-based mesh networks to ensure resilience during outages. Business Strategy: E-commerce platforms should subsidize connectivity costs for businesses and customers in slums, ensuring equitable access to digital platforms. Emerging technologies like blockchain-based payment systems can enhance transaction security and build trust, while AI and machine learning can optimize logistics operations and predictive demand analytics. In the short term, pilot projects can deploy affordable wireless networks and decentralized communication technologies, while long-term strategies involve large-scale investments in telecommunications infrastructure and fintech education programs.

The digital divide's impact on OCM (H1d) and OPD (H2d) further necessitates targeted interventions. Policy Recommendation: Governments and businesses should establish Digital Access Stations (DAS) in high-traffic areas like community centers and marketplaces, providing affordable internet access, devices, and digital literacy training. These stations should be powered by solar energy to address unreliable electricity and feature simplified, multilingual interfaces for low-literacy users. Business Strategy: E-commerce platforms should collaborate with DAS operators to offer incentives such as discounts or free delivery for transactions completed via these stations. Low-tech solutions like SMS-based order and delivery updates can further bridge the digital divide, ensuring inclusivity for users without smartphones. Emerging technologies such as mobile money platforms and blockchain-based financial services can enhance financial inclusion and transaction security, fostering trust and participation in the digital economy. In the short term, pilot projects can test DAS and SMS-based systems, while long-term strategies focus on scaling up digital literacy programs and integrating decentralized fintech solutions.

By explicitly linking these recommendations to the statistical findings, the study provides a robust framework for addressing the infrastructural and technological barriers hindering OCM and OPD in slum communities. For example, the statistical significance of ADS challenges (H1a, H2a) justifies the need for scalable digital addressing solutions, while the impact of poor TPI (H1b, H2b) underscores the importance of localized delivery networks. The integration of emerging technologies, differentiation of short- and long-term solutions, and emphasis on actionable steps ensure that the policy and marketing implications are both practical and impactful. This comprehensive approach not only addresses the immediate challenges but also lays the foundation for sustainable digital inclusion and socio-economic development in underserved areas.

7.2. Theoretical implications

Structural functionalism has not been explicitly used to model online commerce (OCM) and product delivery (OPD) in slum environments. Existing research predominantly focuses on family, education, governance economic, infrastructural, and behavioral challenges without integrating a systemic sociological framework. This study pioneers the application of structural functionalism in this domain, offering a novel perspective that aligns with the interconnected nature of OCM and OPD. Given that slum economies operate through adaptive informal networks, the functionalist approach provides a fitting theoretical lens to examine how infrastructure and technology interact to sustain e-commerce. By framing these elements as interdependent structures within an evolving system, this research extends the functionalist paradigm into an unexplored yet highly relevant technological context.

Unlike traditional technology adoption models such as Technology Acceptance Model (TAM), Unified Theory of Acceptance and Use of Technology (UTAUT), or Diffusion of Innovation (DOI)—which focus on individual-level factors like perceived usefulness, ease of use, and behavioral intentions—this study applies Structural Functionalism (SF) to analyze digital exclusion as a systemic issue. By demonstrating how poor addressing systems, unreliable transportation, weak telecommunications, and digital illiteracy disrupt the e-commerce ecosystem, this study shifts the theoretical perspective from personal adoption barriers to broader structural constraints. This systemic approach challenges the assumption that digital adoption is solely an individual decision and emphasizes the collective role of infrastructural and technological subsystems in enabling or hindering e-commerce participation in slum communities. This perspective is critical in designing structural interventions rather than user-centered solutions, making it a substantial theoretical advancement.

A unique theoretical contribution of this study is the redefinition of OPD as an adaptive mechanism that mitigates systemic inefficiencies in the digital economy. From an SF perspective, OPD serves as a stabilizing force, compensating for infrastructural failures such as inadequate addressing systems, poor transportation networks, and unreliable telecommunication. The study finds that alternative OPD models, such as motorcycle couriers, drone delivery, localized delivery hubs, and

informal logistics networks, act as functional substitutes that restore equilibrium in the e-commerce ecosystem. Unlike conventional logistics theories, which emphasize supply chain optimization in structured markets, this study highlights adaptive delivery mechanisms in resource-constrained environments, providing new insights into last-mile delivery strategies in slum economies. This novel recontextualization of OPD broadens the scope of logistics research by demonstrating how alternative delivery models compensate for systemic dysfunctions and ensure e-commerce functionality in underserved areas.

A key limitation of existing digital inclusion theories is their tendency to treat infrastructure and technology as separate domains. This study bridges this gap by demonstrating how infrastructural barriers (e.g., poor road networks, unreliable telecommunications) and technological barriers (e.g., low digital literacy, limited e-payment adoption) are deeply interconnected. For instance, weak telecommunications infrastructure not only disrupts online transactions but also reinforces the digital divide by restricting access to digital platforms. Unlike technology acceptance theories, which assume that users make rational choices based on technological attributes, this study emphasizes how systemic failures create a self-reinforcing cycle of digital exclusion. By framing these challenges as interdependent dysfunctions, this study proposes a new theoretical framework that integrates infrastructure and technological capacity as co-dependent variables in shaping digital inclusion. This systemic perspective is crucial in designing integrated interventions, rather than fragmented solutions, for digital participation in slum environments.

Most e-commerce adoption theories focus on individual-level trust (e.g., trust in online platforms, payment systems, and sellers) from a security concern perspective. However, this study introduces security concerns (trust) as a systemic factor within the SF framework, demonstrating that trust deficits are not just individual perceptions but structural imbalances that weaken the e-commerce ecosystem. Findings reveal that security concerns (e.g., online fraud, data breaches, physical risks to delivery personnel) disrupt the interdependent trust network, preventing both consumers and businesses from engaging in digital commerce. This theoretical innovation highlights that trust must be understood not only as a behavioral factor but also as a structural condition dependent on the reliability of financial, technological, and logistical subsystems. By framing trust as a cohesive mechanism that maintains digital marketplace equilibrium, this study extends Structural Functionalism to digital commerce research, offering new pathways for strengthening consumer confidence in slum-based OCM and OPD.

This study makes a significant theoretical contribution by highlighting the limitations of generalizability in mainstream e-commerce and logistics management theories, particularly when applied to slum-based economies. Existing technology adoption and logistics models are predominantly built on formal urban infrastructures, assuming standardized addressing systems, developed transportation networks, stable telecommunications, and digital financial inclusion. However, these assumptions do not hold in slum environments, where infrastructural and socio-economic challenges significantly shape OCM and OPD. The study's findings indicate that slum-based OCM and OPD face distinct barriers, such as inadequate ADS, poor TPI, UTC, DTD, low EDL, and STC. These factors create unique e-commerce and logistical constraints that cannot be fully explained by traditional e-commerce models such as TAM, UTAUT, or DOI theory. This necessitates the development of a slum-specific theoretical framework that integrates alternative payment solutions, informal delivery networks, and digital literacy programs into the e-commerce and logistics discourse. By identifying the limitations of existing frameworks, this study provides a foundation for developing new theoretical models that address the realities of slum-based digital transactions and logistics operations. Future research can build upon this by incorporating institutional support mechanisms, last-mile delivery optimization strategies, and community-driven fintech solutions, paving the way for inclusive digital commerce ecosystems in slum communities.

The novel findings of this study highlight the critical role of emerging technologies in addressing the infrastructural and technological barriers hindering OCM and OPD in SSA slums. Traditional e-commerce models fail to account for informal ADS, unreliable TPI, and UTC, but technological advancements offer transformative solutions. One of the major barriers identified is the absence of formal ADS, which disrupts last-mile delivery efficiency. Innovative solutions such as What3Words, Google Plus Codes, and AI-driven geospatial mapping provide precise digital addresses, enabling logistics providers to navigate complex slum environments more effectively. Additionally, mobile money platforms (e.g., M-Pesa, Flutterwave) and blockchain-based financial services enhance transaction security, financial inclusion, and trust, reducing fraud risks and offering viable alternatives to cash-based transactions. Furthermore, AI and machine learning optimize logistics operations, predictive demand analytics, and supply chain management, while drone logistics, electric bikes, and decentralized crowd-sourced delivery networks provide cost-effective OPD solutions in underserved communities. The study also emphasizes the importance of EDL and fintech education, advocating for mobile-based training programs to empower slum residents in e-commerce. By unveiling these novel insights, this study establishes emerging technologies as critical enablers of slum-based OCM and OPD, offering practical solutions that can transform digital transactions and logistics operations in underserved communities. By integrating these innovations, slum-based OCM and OPD can overcome existing barriers, fostering greater economic inclusion and enabling the development of sustainable digital commerce ecosystems in SSA slums.

7.3. Limitations and future directions

This study offers valuable insights into the factors shaping the participation of SSA slum residents and businesses in the digital economy, particularly in enhancing e-commerce and product delivery. However, certain limitations must be acknowledged to contextualize the findings and provide direction for future research. A key limitation lies in the study's focus on infrastructural and technological factors, which, while crucial, do not capture the full spectrum of influences on online commerce and product delivery within slum environments. Social and cultural dimensions, such as community trust, informal economic practices, and collective decision-making, play a significant role in shaping technology adoption and digital transactions. The exclusion of these factors may result in an incomplete representation of the complexities within slum economies, limiting the applicability of the findings beyond the immediate scope of infrastructure and technology. Addressing these additional influences in future studies could provide a more comprehensive understanding of digital engagement in slum settings. Additionally, the geographic scope of the study is confined to SSA slums, which presents challenges in applying the findings to slum communities in other regions such as South Asia or Latin America. While SSA slums exhibit distinct infrastructural and economic characteristics, other regions may face unique socio-cultural and economic conditions that influence e-commerce participation differently. For instance, hierarchical social structures in South Asia or closer formal economic integration in Latin America may create distinct barriers or facilitators to digital commerce. The regional specificity of the study underscores the need for comparative research across different slum environments to establish broader applicability.

The methodological approach further presents a limitation, as the study relies exclusively on quantitative methods. While statistical analyses offer critical insights into the relationships between infrastructural and technological factors and e-commerce participation, they do not capture the depth of lived experiences, behavioral nuances, or contextual barriers that qualitative approaches could reveal. Perspectives on trust in digital platforms, resistance to formalized online transactions, and policy effectiveness in digital inclusion efforts are better explored through qualitative methods such as interviews, focus groups, or ethnographic studies. Integrating qualitative approaches in future

research would provide a richer and more contextualized understanding of the barriers and opportunities within slum-based digital commerce. To build on the findings of this study, future research should expand the scope of influencing variables beyond infrastructure and technology to include socioeconomic, cultural, and policy-related factors. A mixed-methods approach that combines quantitative data with qualitative insights could enhance the depth of analysis and provide a more holistic view of e-commerce participation in slum settings. Moreover, broadening the geographic focus to include slums in other regions would enable comparative analyses and strengthen the generalizability of the findings. These directions would contribute to a more comprehensive understanding of digital inclusion and e-commerce growth including product delivery in underserved communities globally.

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CRediT authorship contribution statement

Abdul Karim Armah: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jinfa Li:** Writing – review & editing, Supervision, Resources, Funding acquisition. **Mengdi Wei:** Visualization, Software, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Table 2
The empirical values of the instrumental and quadratic variables

Instrumental variable	Eigen values	Proportion explained	Initial variables	Eigen vectors	Correlation coefficient	Bartlett test (pValue)	KMO sampling adequacy
OPD_IV	1.3443	0.6611	OPD1	0.5488	0.6159	0.0000	0.7761
	1.3002		OPD2	0.1085	0.6908		
			OPD3	0.6392	0.4794		
			OPD4	0.1155	0.5589		
STC_IV	1.4644	0.6299	STC1	0.5281	0.6991	0.0000	0.7143
	1.0551		STC4	0.4548	0.4939		
			STC2	0.6688	0.4353		
			STC3	0.6633	0.4289		
ADS_IV	1.6122	0.6724	ADS1	0.4165	0.4010	0.0000	0.7369
	1.0773		ADS2	0.3153	0.7110		
			ADS3	0.0748	0.5943		
			ADS4	0.6511	0.6405		
DTD_IV	1.3186	0.6434	DTD4	0.7020	0.3250	0.0000	0.6040
	1.2554		DTD2	0.6300	0.3080		
			DTD1	0.6480	0.3130		
			DTD3	0.5950	0.3000		
STCsq	1.3463	0.61420	STC1	0.7140	0.3440	0.0000	0.6760
	1.1107		STC4	0.5710	0.3080		
			STC2	0.5450	0.3000		
			STC3	0.6280	0.3220		
DTDsq	1.4005	0.6562	DTD4	0.7040	0.3200	0.0000	0.6850
	1.2245		DTD2	0.6430	0.3060		
			DTD1	0.6590	0.3090		
			DTD3	0.6190	0.3000		
ADSsq	1.3387	0.6571	ADS1	0.6710	0.3120	0.0000	07350
	1.2893		ADS2	0.6590	03090		
			ADS3	0.7320	0.3260		
			ADS4	0.5660	0.2860		

The Kaiser-Meyer-Olkin (KMO) index of sampling adequacy for OPD_IV, STC_IV, ADS_IV, DTD_IV, STCsq, DTDsq and ADSsq are 0.7761, 0.7143, 0.7369, 0.6040, 0.6760, 0.6850 and 07350 respectively, which is acceptable and supports the reliability of our sampled data.

Table 3
The empirical values of unobserved heterogeneity model criteria

Segment	1	2	3	4	5	6
AIC	66586.529	28823.219	21392.663	17947.813	14705.915	**12820.309**
AIC3	66602.529	28856.219	21442.663	18014.813	14789.915	**12921.309**
CAIC	66709.261	29076.355	21776.202	**13595.057**	18461.755	15350.259
AIC4	66618.529	**13022.309**	21492.663	18081.813	14873.915	28889.219
BIC	66693.261	29043.355	21726.202	**13494.057**	18394.755	15266.259
MDL5	67248.19	30187.897	23460.356	20718.522	18179.639	**16997.049**
EN	**0**	0.992	0.938	0.938	0.867	0.816

Table 9
The wording used for respective items in slum OCM and OPD.

Construct	Survey Items	Operational Definitions
Digital Divide (DTD)	DTD1	Access to the internet is a major challenge for slum residents that engage in online business and shopping.
	DTD2	Technologies for digital services are limited to slum businesses and residents.
	DTD3	Slum residents struggle to navigate online platforms.
	DTD4	The digital gap prevents equal participation in the benefits of online commerce.
Education and Digital Literacy (EDL)	EDL1	The lack of education in slum areas makes it difficult to participate in online business transactions.
	EDL2	Residents of the slum have a low awareness of internet commerce platforms.
	EDL3	Digital literacy skills are scarce in the slums, making it difficult for individuals to engage in online commerce.
	EDL4	In slum region, there are insufficient educational initiatives to help residents improve their computer abilities.
Addressing System (ADS)	ADS1	The addressing system in slum areas is unreliable for online product deliveries.
	ADS2	Providing precise address information for online transactions is often problematic.
	ADS3	Product delivery agents normally struggle to locate residence in communities when delivering online orders.
	ADS4	The absence of consistent addressing makes it difficult to receive items on time.
Transportation Infrastructure (TPI)	TPI1	Transportation issues in slum communities regularly affect the timely delivery of online ordered products.
	TPI2	Poor road conditions contribute to delays in receiving online purchases.
	TPI3	Delivery services are frequently disrupted owing to poor transportation infrastructure.
	TPI4	Improving transportation infrastructure in the slums will considerably increase the efficiency of online goods delivery.
Unreliable Telecommunication (UTC)	UTC1	Unstable internet connectivity in slum communities hinders online commerce and product delivery activities.
	UTC2	The lack of reliable telecommunication services makes it challenging to use navigation apps to optimize delivery routes and goods tracking.
	UTC3	Slow internet speeds negatively impact online shopping experience and resolving issues with online orders.
	UTC4	Often experience dropped calls or poor reception when communicating with online vendors and delivery personnel.
Security Concerns (STC)	STC1	Concerned about the safety of personal information when making online purchases.
	STC2	Congestion in the slum makes the environment prone to theft and pose danger to delivery personnel.
	STC3	The risk of online fraud makes me hesitant to shop online frequently.
	STC4	The risk of online fraud makes me hesitant to shop online frequently.
Online Commerce (OCM)	OCM1	Online commerce platforms are accessible to slum residents.
	OCM2	Purchasing goods online is convenient for slum dwellers.
	OCM3	Online commerce offers a wide range of products ideal for slum dwellers.
	OCM4	Slum residents find online commerce platforms to be user-friendly.
Online Product Delivery (OPD)	OPD1	Online product delivery services are reliable in slum communities.
	OPD2	Slum customers are satisfied with the effectiveness of online product delivery.
	OPD3	Using collection and delivery points in slum communities minimizes delivery challenges.
	OPD4	The availability of tracking systems in online product deliveries boosts the confidence and trust of slum residents.

Data availability

Data will be made available on request.

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